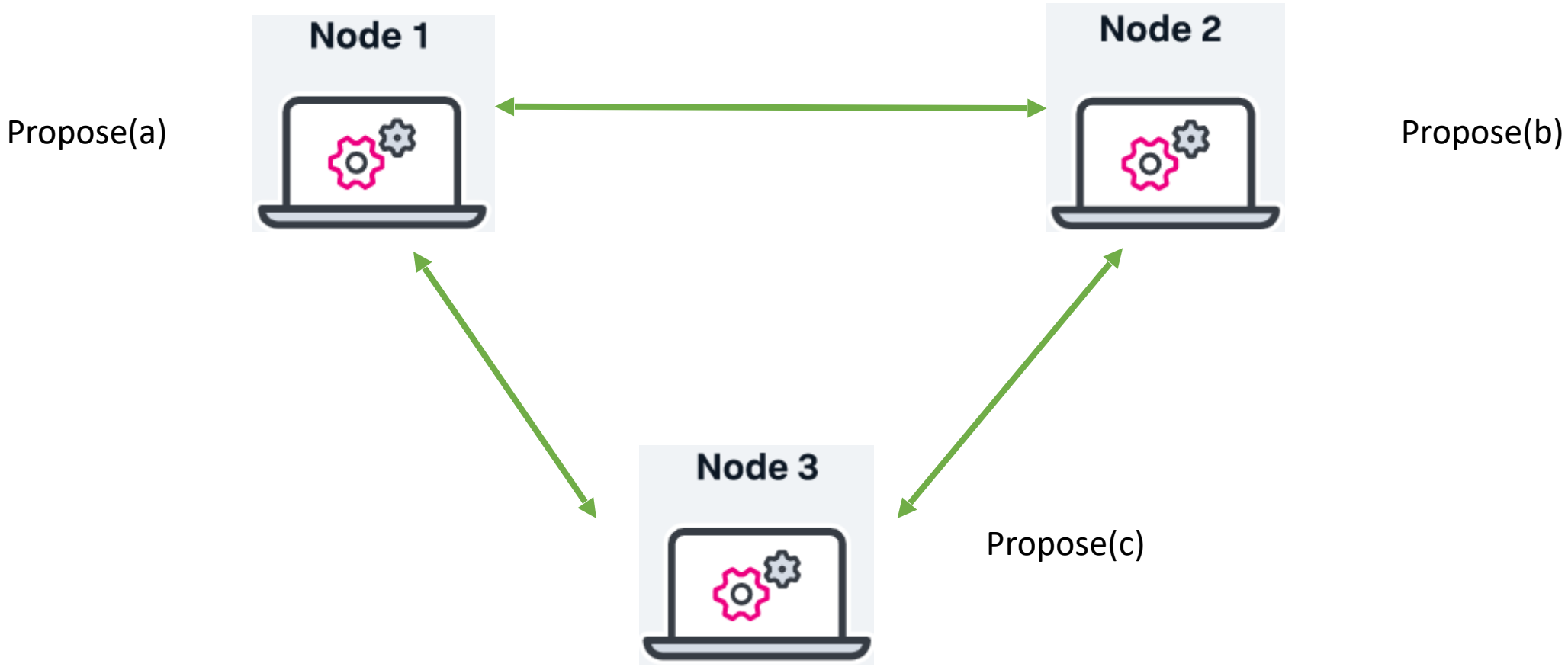
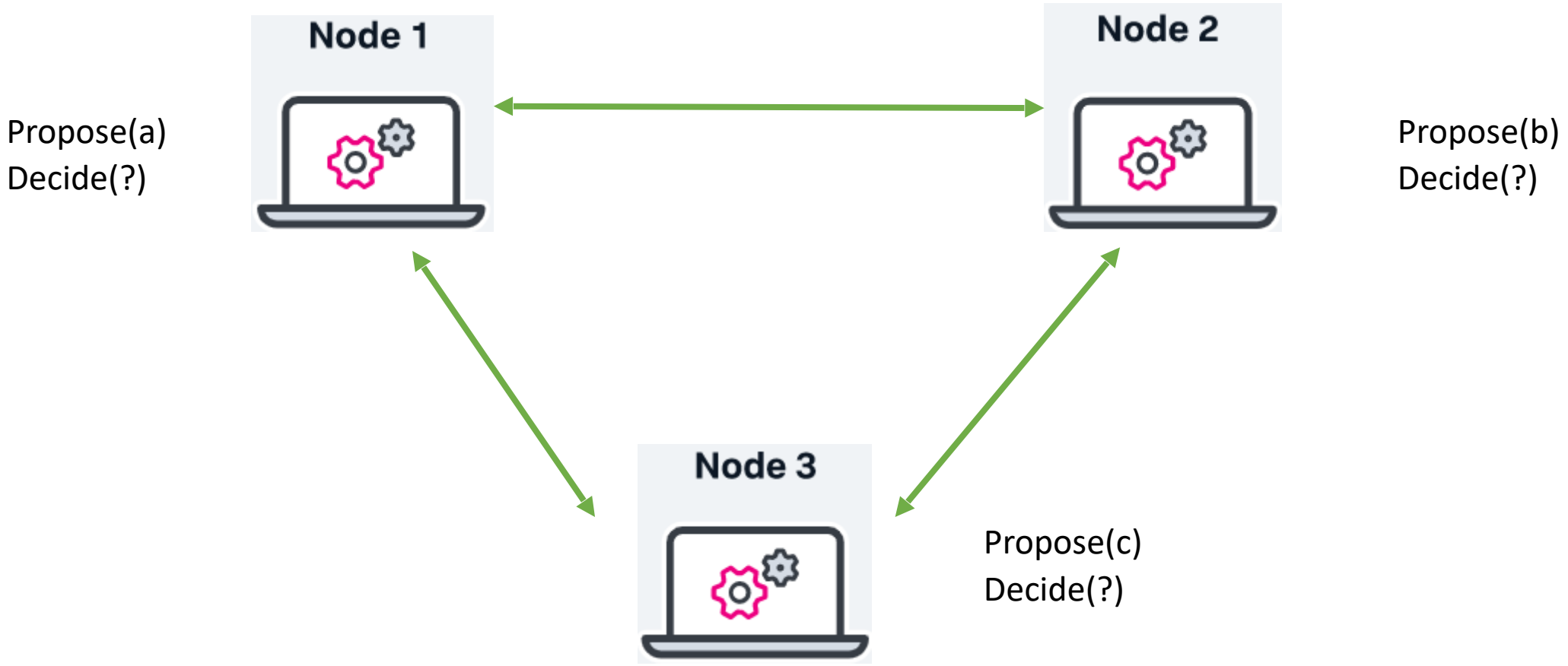
Consensus

Mohsen Lesani

Consensus



Consensus


Node 1
Node 2
Node 3
Propose(a)
Decide(?)
Propose(b)
Decide(?)
Propose(c)
Decide(?)

Consensus

In the consensus problem, the processes propose values and have to agree on one among these values.

Solving consensus is key to solving many problems in distributed computing (e.g., total order broadcast, atomic commit, terminating reliable broadcast and replicated services).

Consensus

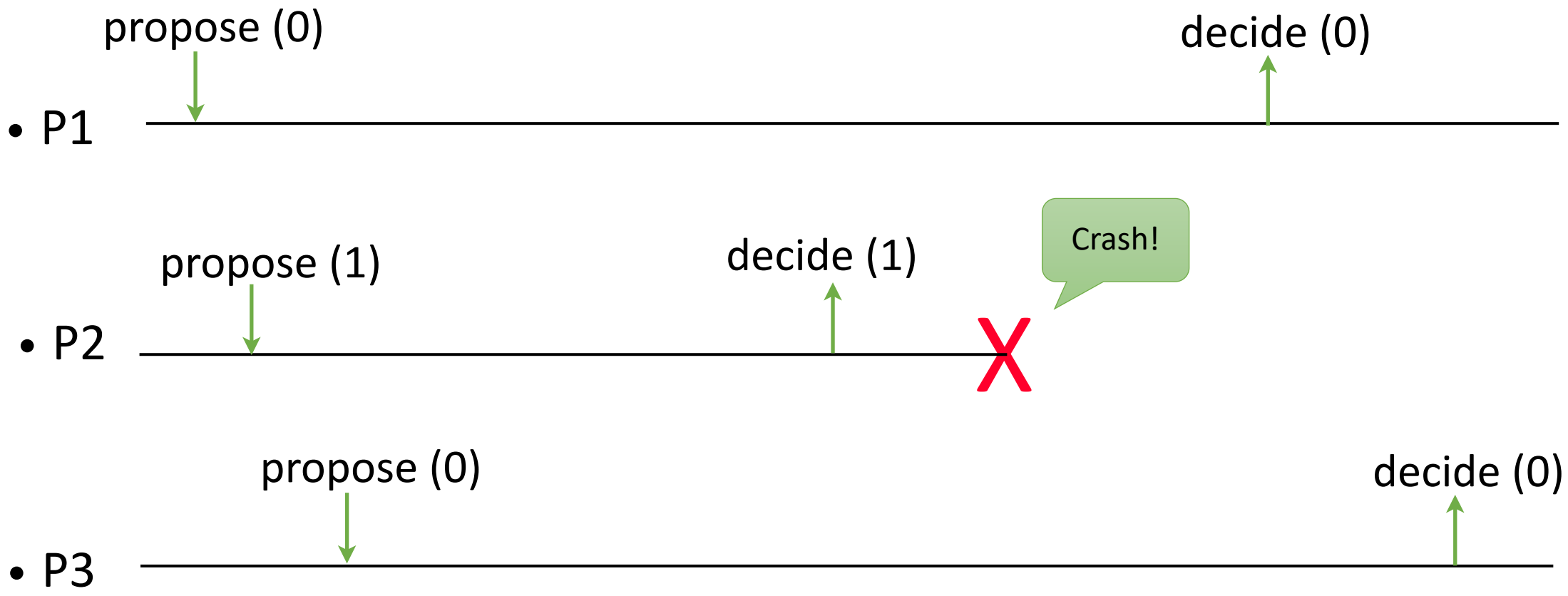
- **Events**
  - Request: < propose(v) >
  - Indication: < decide(v) >

- **Properties:**
  - **C1, C2, C3, C4**

Consensus

- **C1. Validity**: Any value decided is a value proposed.
- **C2. Agreement**: No two **correct** processes decide differently.
- **C3. Termination**: Every correct process eventually decides.
- **C4. Integrity**: No process decides twice.

Consensus



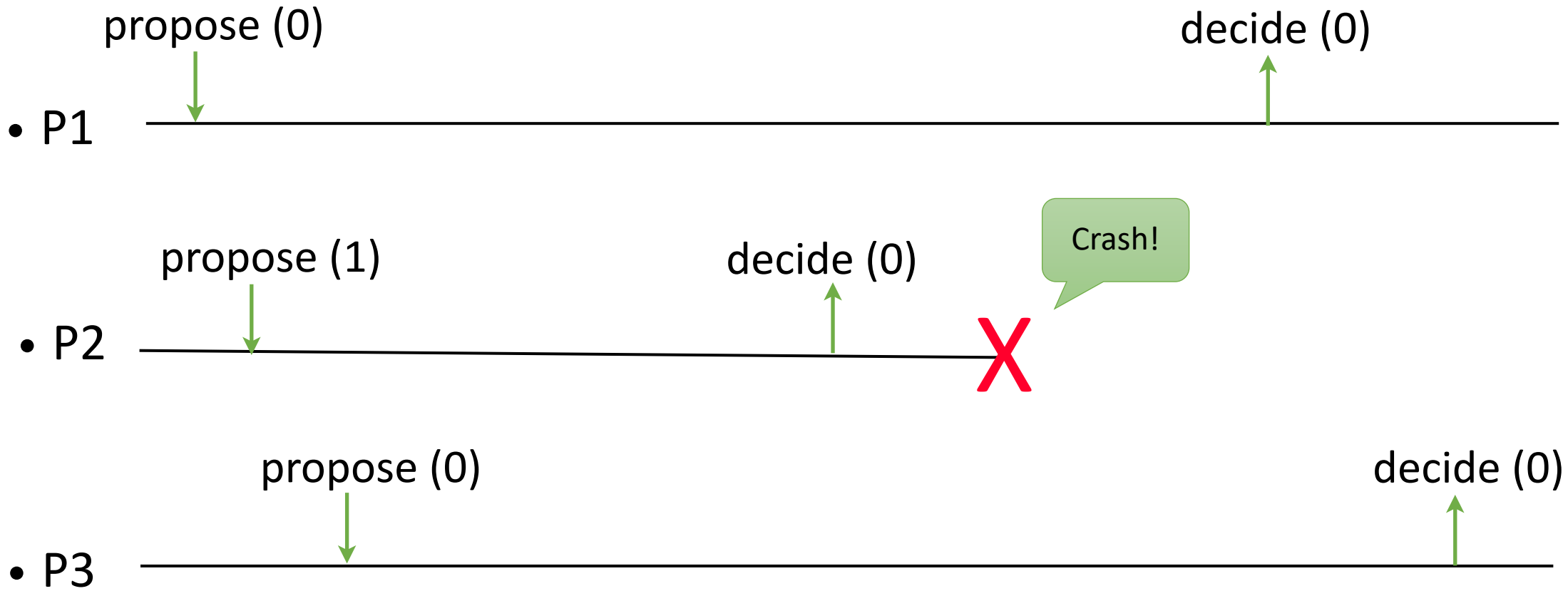
Correct Execution. Correct processes decide the same value.

Uniform Consensus

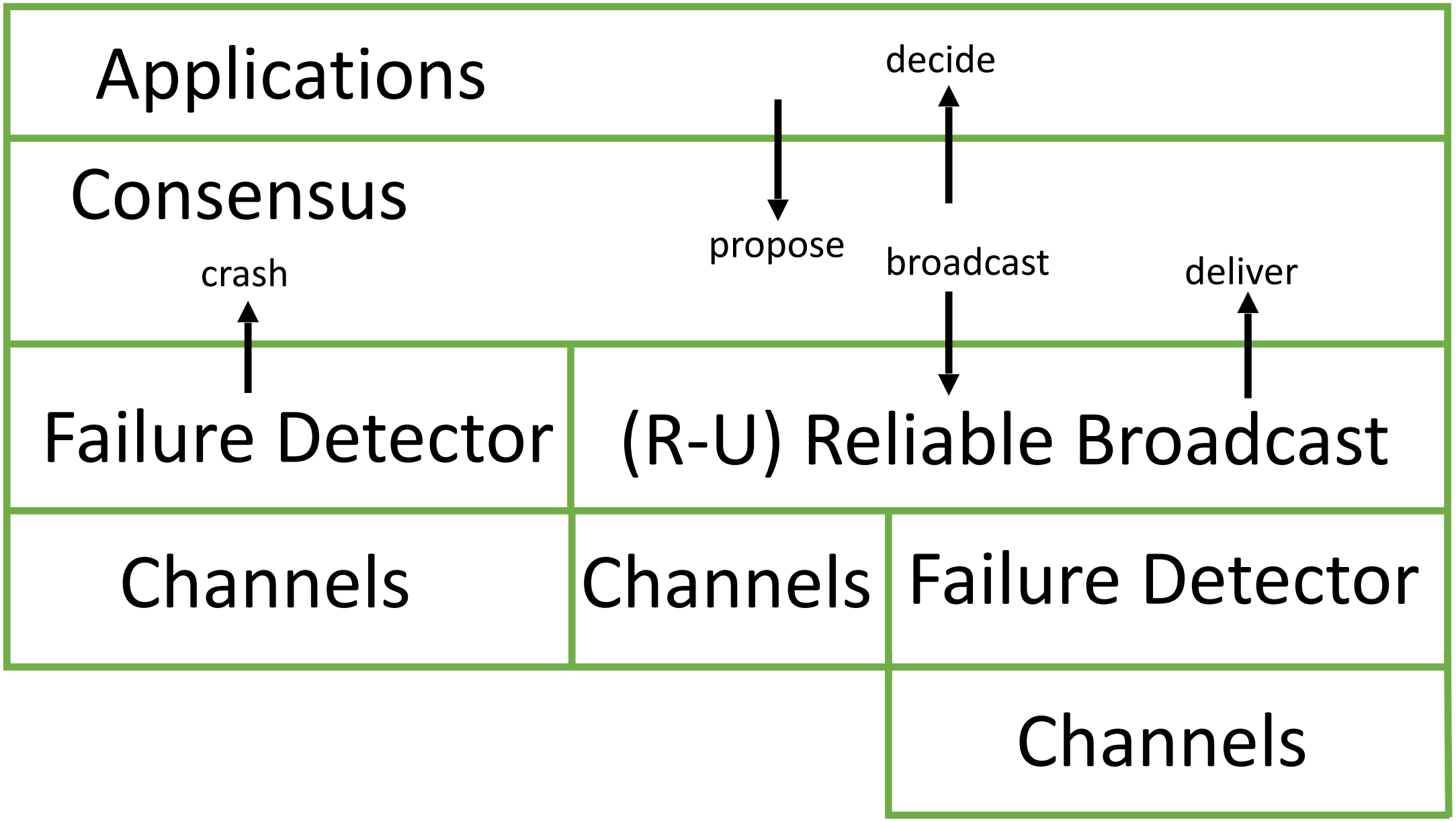
- **C1. Validity**: Any value decided is a value proposed.
- **C2’. Uniform Agreement**: No two **processes** decide differently.
- **C3. Termination**: Every correct process eventually decides.
- **C4. Integrity**: No process decides twice.

Even incorrect processes should not decide differently.

Uniform Consensus



Modules of a process



Three algorithms

- A P-based (i.e., fail-stop) consensus algorithm
- A P-based (i.e., fail-stop) uniform consensus algorithm
- A <>P-based (i.e. fail-silent) uniform consensus algorithm assuming a correct majority

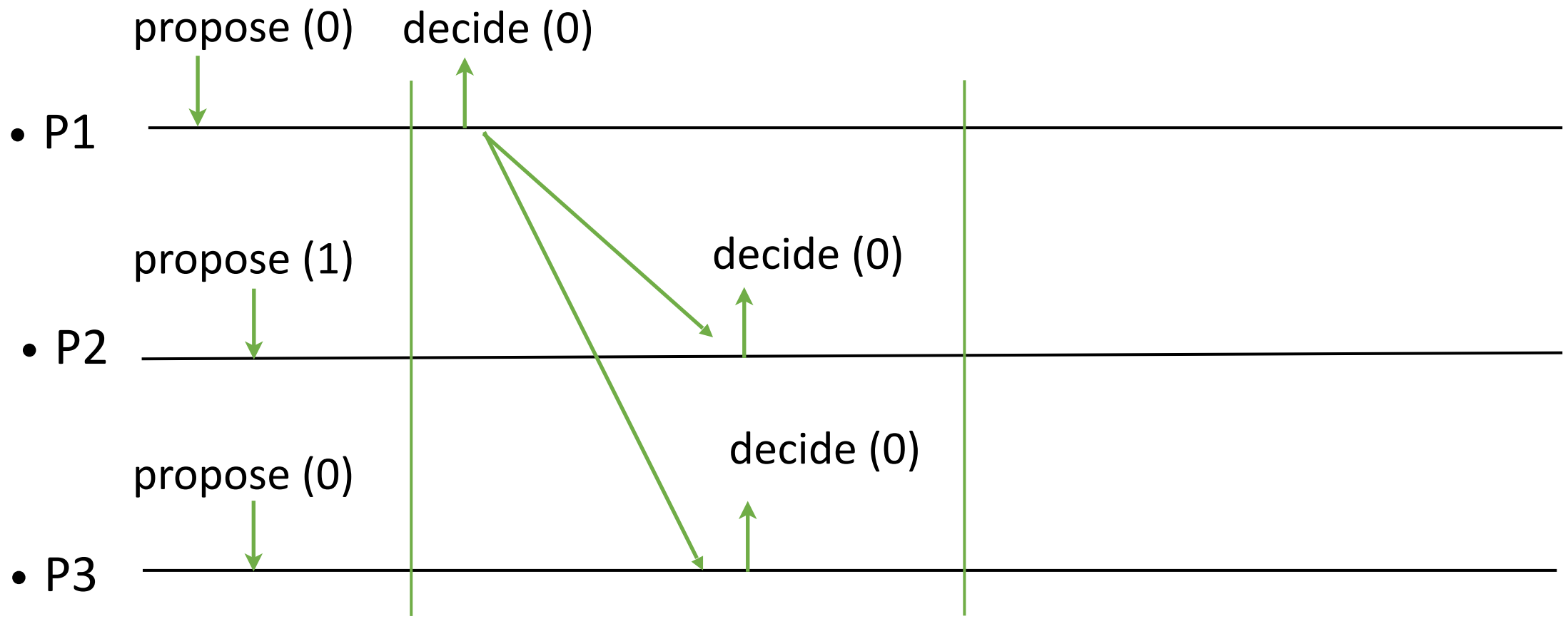
P is the perfect failure detector.
<>P is the eventually perfect failure detector.

Fail-stop Consensus

- A P-based (i.e., fail-stop) consensus algorithm

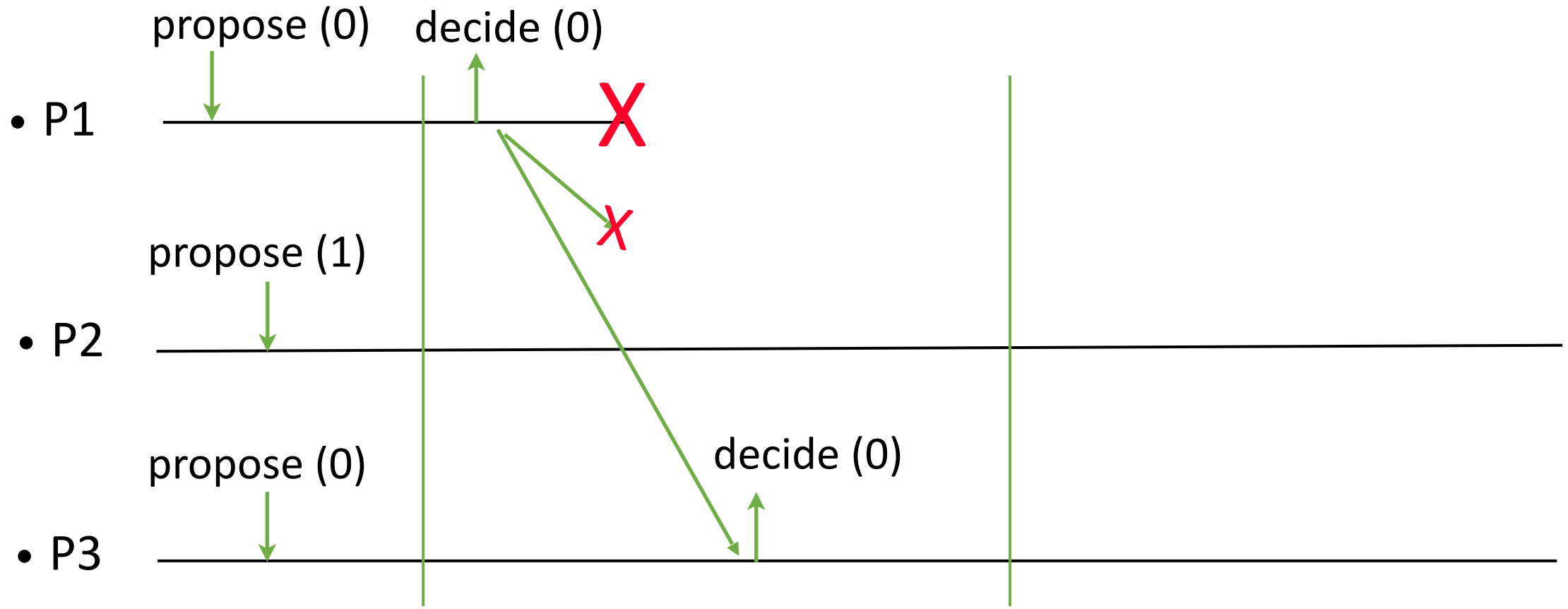
P is the perfect failure detector.

Fail-stop Consensus



Can the first process decide its own value and impose it on others?

Fail-stop Consensus

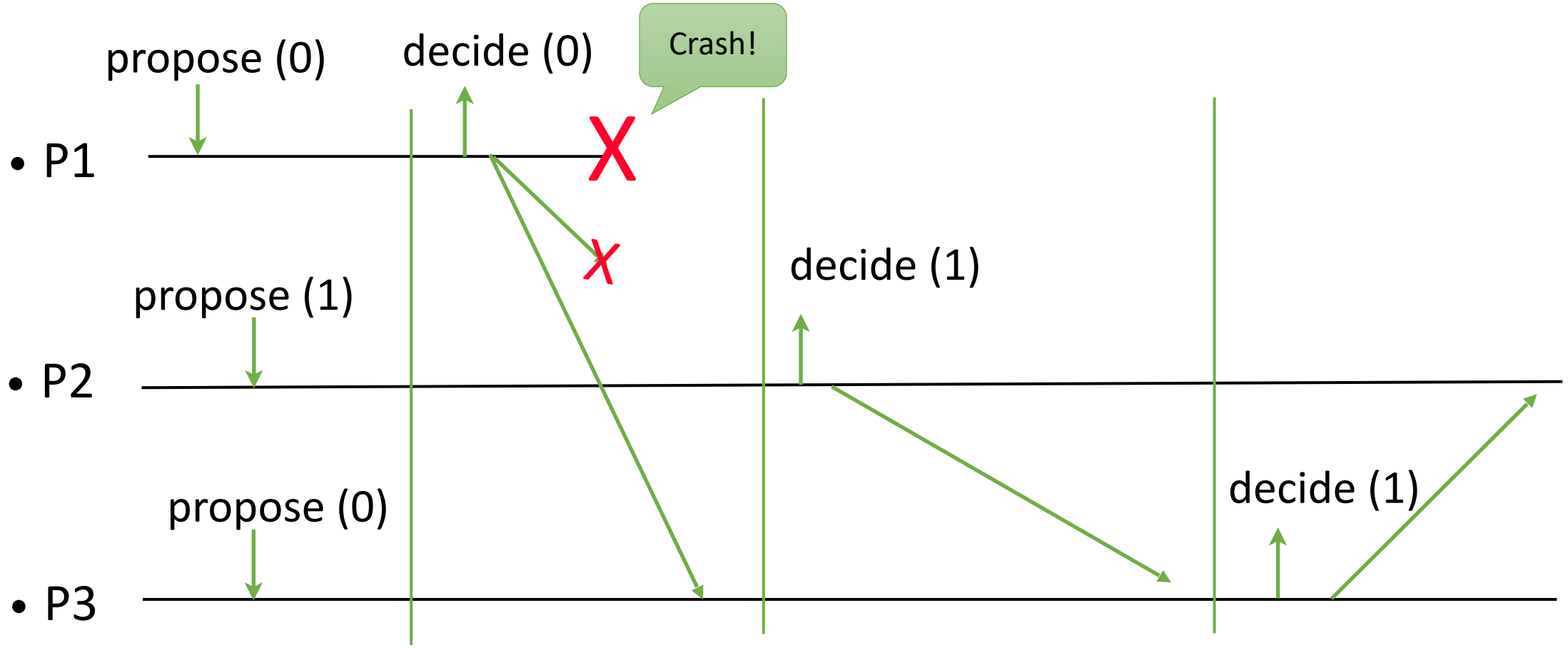


What if he crashes? P2 never decides.

Fail-stop Consensus

Idea:
Given an order for the processes, the first **correct** process broadcasts its proposal and imposes it on others.

Fail-stop Consensus

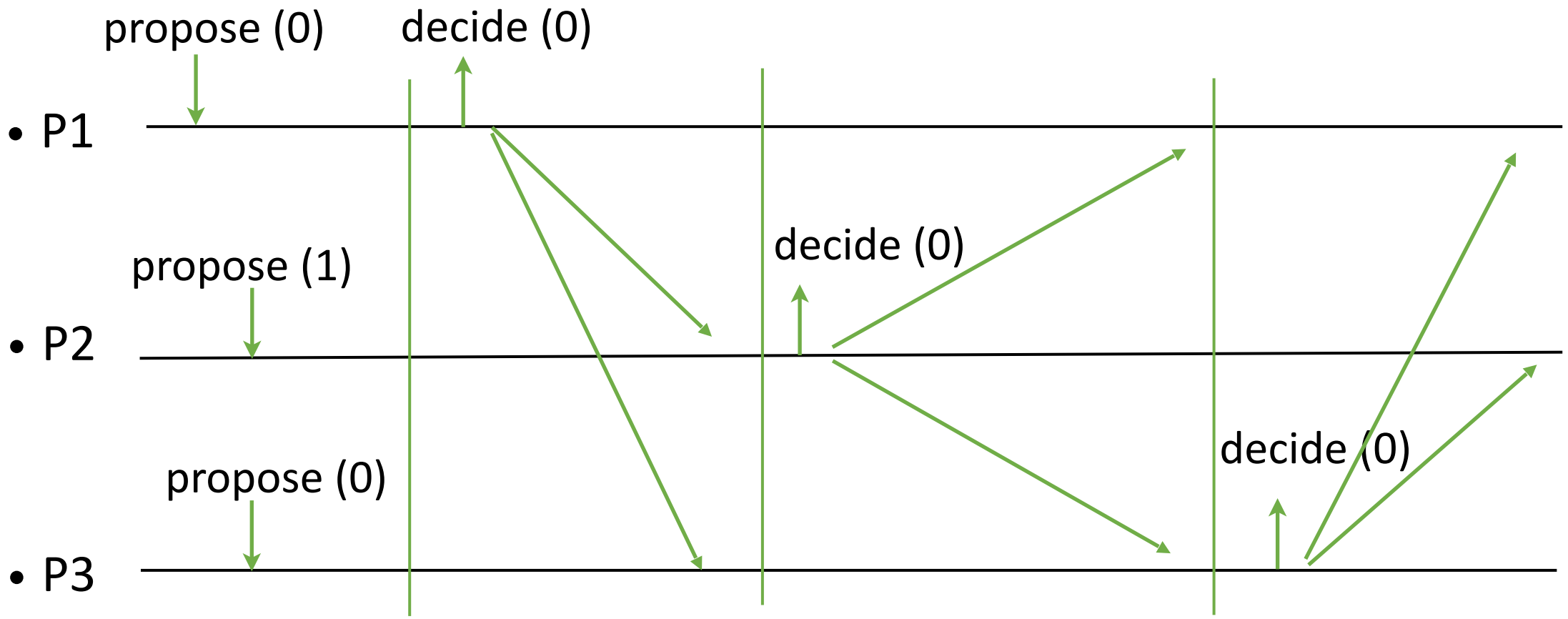


We note that p3 cannot decide in round 1 because p1 might crash and not impose the value on p2.
Then p2 can decide differently.

Fail-stop Consensus

- The processes go through rounds incrementally (1 to n): in each round, the process with the id corresponding to that round is the leader of the round.
- The leader of a round decides its current proposal and broadcasts it to all.
- A process that is not leader in a round waits to
  (a) deliver the proposal of the leader in that round and adopts it, or
  (b) suspect the leader

Fail-stop Consensus



The first correct leader propagates its value throughout the system. Later leaders go through the later rounds because they don't know whether there has been a correct leader before them. The leader of a round could be the first correct leader.

Fail-stop Consensus

Implements: Consensus (cons).

Uses:

BestEffortBroadcast (beb).

PerfectFailureDetector (P).

upon event < Init > **do**

suspected := ∅

round := 1; prop := ⊥

broadcast := delivered[] := false

upon event < propose(v) > **do**

if prop = ⊥ **then**

prop := v

upon event < P, crash(pi) > **do**

suspected := suspected U {pi}

delivered map is used to remember if a message is delivered in a round.

broadcast is used to not broadcast twice.

Fail-stop Consensus

upon event p_{round}=self **and** prop ≠⊥ **and** broadcast=false **do**

trigger <decide(prop)>

trigger <beb, broadcast(prop)>

broadcast := true

When it is my turn, I decide my current proposal and broadcast it.

Fail-stop Consensus

upon event < beb, deliver(p_{round}, value) > **do**

    prop := value

    delivered[round] := true

upon event delivered[round] = true **or** $p_{round} \in$ suspected **do**

    round := round + 1

Correctness argument

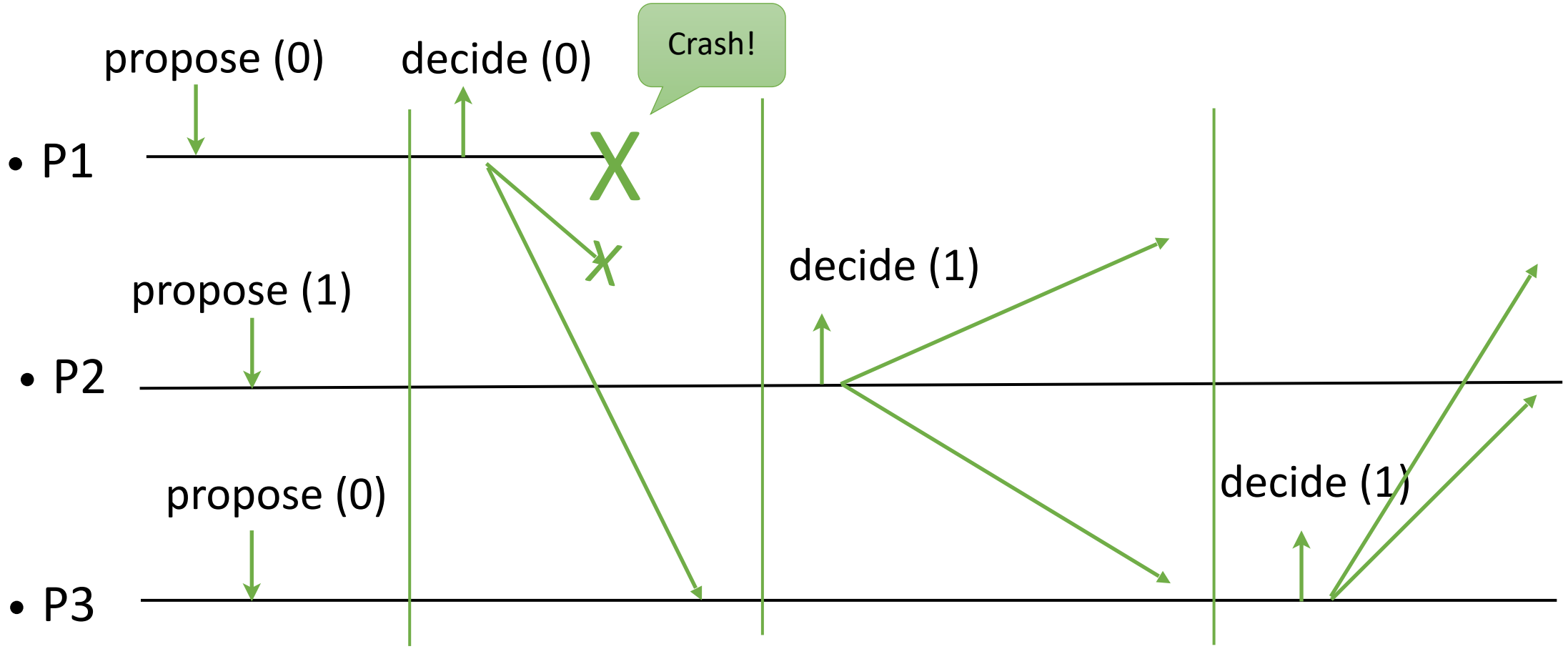
Agreement:

- Let pi be the correct process with the smallest id.
- Assume pi decides v.
- In round i, all correct processes receive and adopt v. From that round on, no other value exists in the system, and every correct process eventually decides v.

Fail-stop Uniform Consensus

A P-based (i.e., fail-stop) **uniform** consensus algorithm

Non-uniformity in the previous protocol

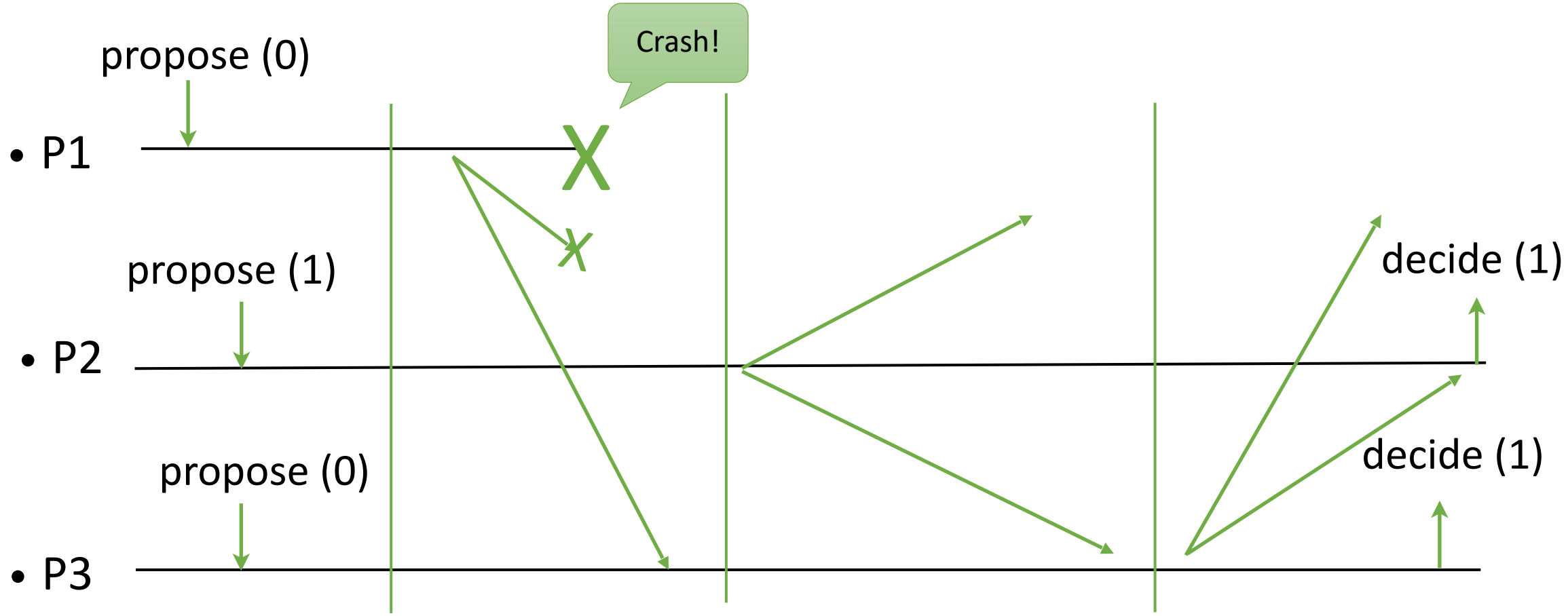


Fail-stop Uniform Consensus

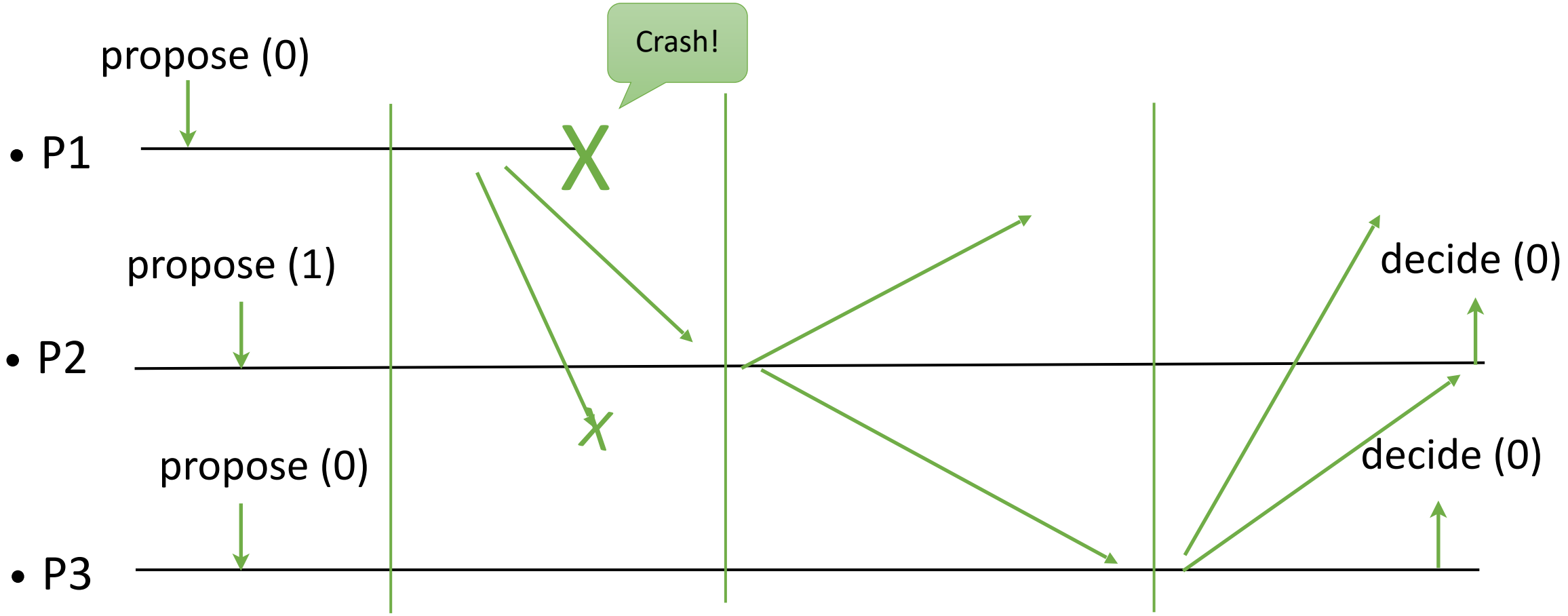
The idea:

- Not letting a process decide and then crash before imposing its value on everyone.
- So we delay the decision.
- The first correct process that succeeds imposing its value may be the last one. Therefore, we delay the decision to the last round.
- The processes exchange and update their proposals in rounds, and after n rounds decide on the current proposal value.

Fail-stop Uniform Consensus

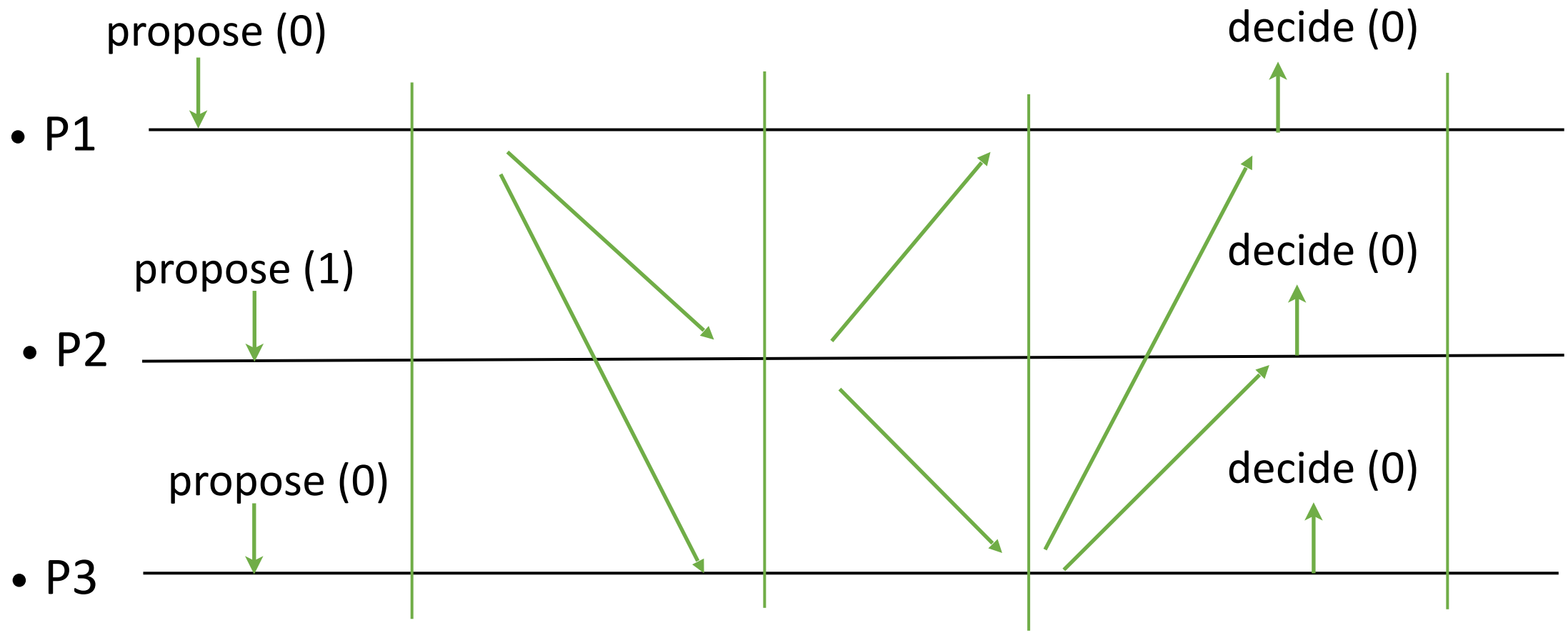


Fail-stop Uniform Consensus



The proposal of the crashed process is decided.

Fail-stop Uniform Consensus



Fail-stop Uniform Consensus

- The processes go through rounds incrementally (1 to n).
- In each round i, process pi sends its current proposal to all.
- A process adopts any proposal it receives.
- Processes decide on their current proposal values at the end of round n.

Fail-stop Uniform Consensus

Implements: Uniform Consensus (ucons).

Uses:

BestEffortBroadcast (beb).

PerfectFailureDetector (P).

upon event < Init > **do**

suspected := ∅

round := 1; prop :=⊥

broadcast := delivered[] := false

decided := false

upon event < propose(v) > **do**

if prop = ⊥ **then**

prop := v

upon event < P, crash(pi) > **do**

suspected := suspected U {pi}

decided is used to not decide twice.

Fail-stop Uniform Consensus

upon event p_{round}=self **and** prop ≠⊥ **and** broadcast=false **do**

trigger <beb, broadcast(prop)>

broadcast := true

Fail-stop Uniform Consensus

upon event < beb, deliver(p_{round}, value) > **do**

    prop := value

    delivered[round] := true

upon event delivered[round] = true **or** $p_{round} \in$ suspected **do**

    if round=n **and** decided=false **then**

        trigger <decide(prop)>

        decided := true

    else

        round := round + 1

Correctness argument

Uniform agreement:

- Consider the *decided process* p_i *with the lowest id*.
- Consider the processes p_j that have not crashed by round i. At round i, p_j must have adopted the proposal of pi or suspected pi.
- By the accuracy property of P, p_i could not be suspected at round i. Thus, p_j has adopted the proposal of p_i at round i.
- Since then, that value has been the only value in the system.
- This includes the values of the processes in round n which they decide.

Fail-silent Consensus

A <>P-based uniform consensus algorithm assuming a correct majority.

<>P is the eventually perfect failure detector.

Leader-driven consensus

- The most important paradigm.
- Introduced (as total-order broadcast) in
  - Viewstamped replication
  - Paxos
  - PBFT
- It is used in many cloud service platforms.

<>P

<>P ensures:

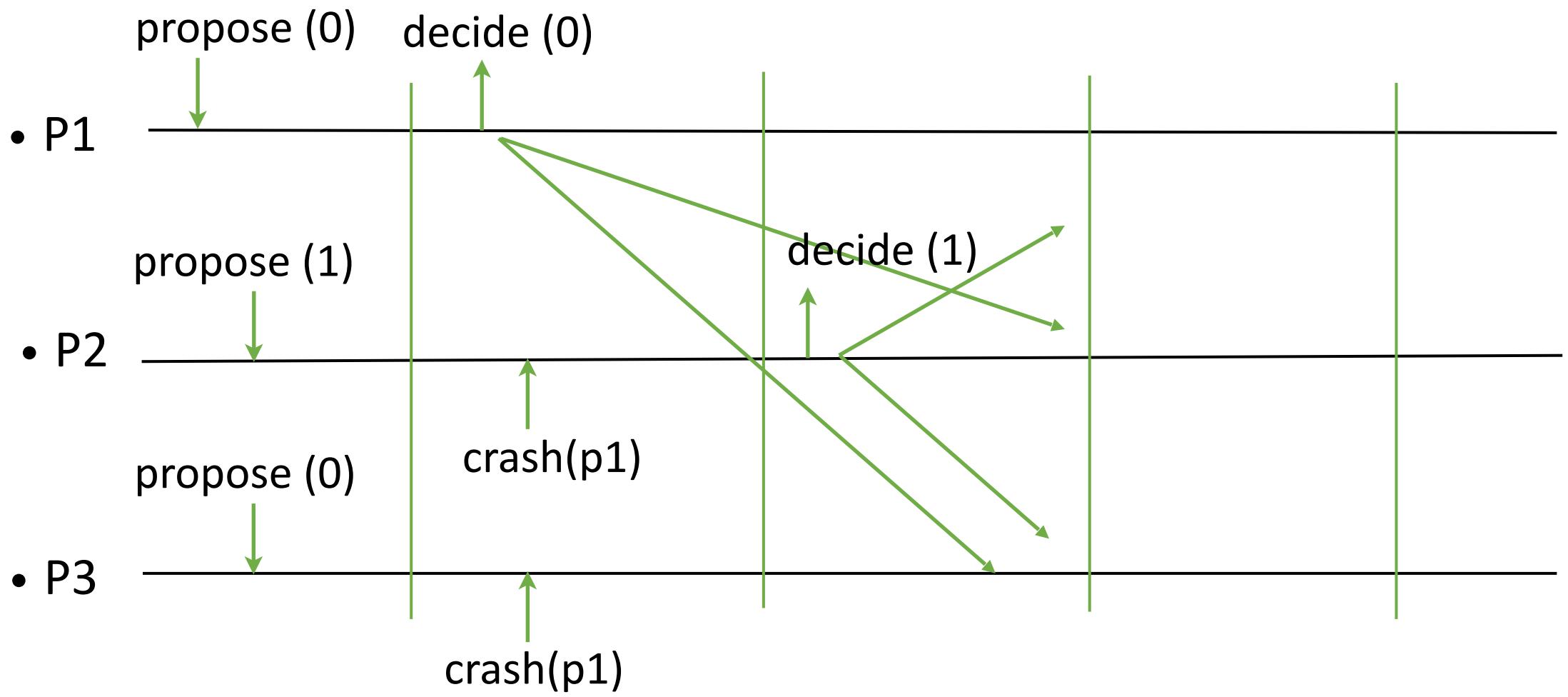
- **Strong completeness**: Eventually, every process that crashes is permanently suspected by all correct processes.
- **Eventual strong accuracy**: Eventually, no correct process is suspected by any process.

<>P

“<>” makes a difference:

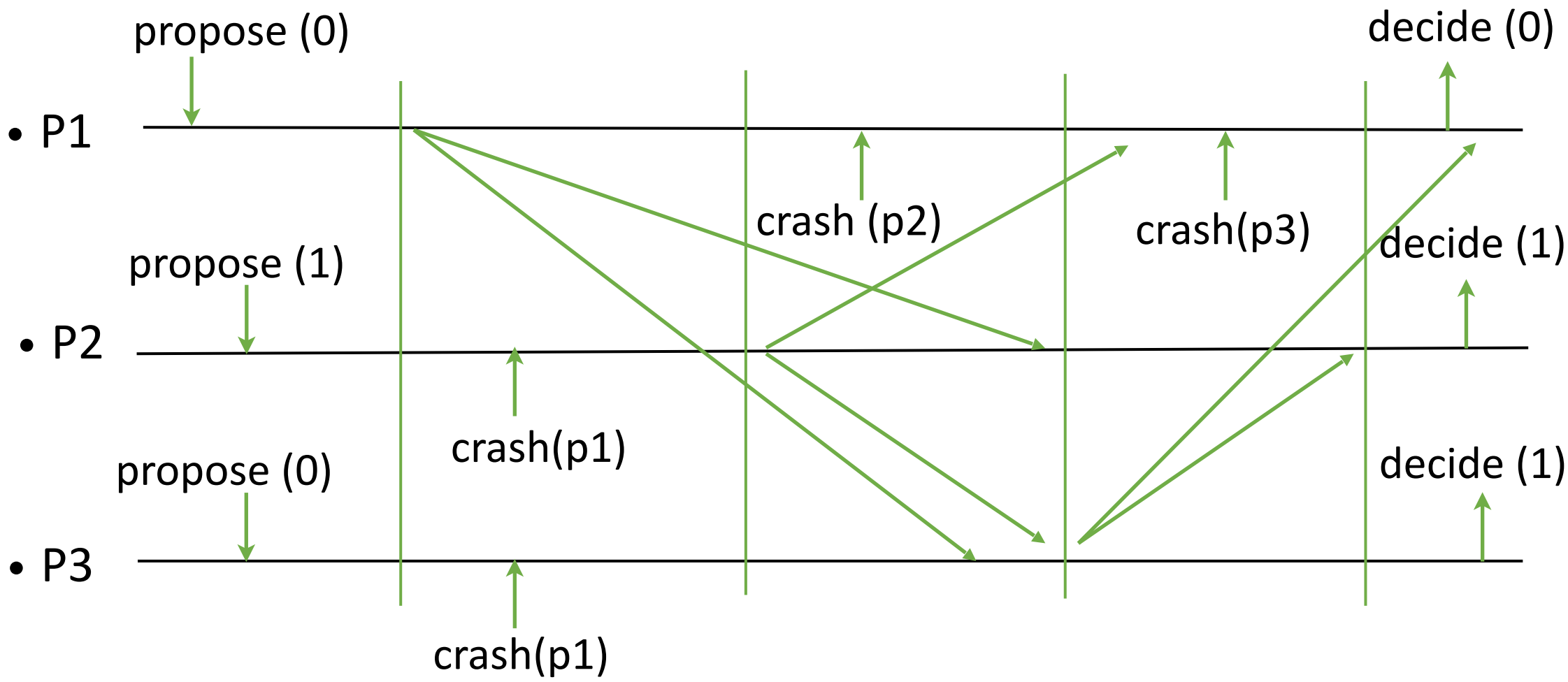
- **Eventual strong accuracy:** strong accuracy holds only after finite time.
- Correct processes may be falsely suspected a finite number of times.
- This breaks consensus algorithms I and II.

Agreement violated with <>P in protocol 1



The processes p2 and p3 move to the next round too early. Process P2 decides 1. Agreement is violated.

Agreement violated with <>P in protocol 2



Inaccurate crash messages make P2 and P3 miss the proposal value from P1, and similarly P1 misses proposal values from P2 and P3.

Fail-silent Consensus

The idea:

A correct leader tells processes what to decide.

A leaders can fail or be falsely suspected to have failed. It should change. To preserve agreement, it leaves a trace for the next.

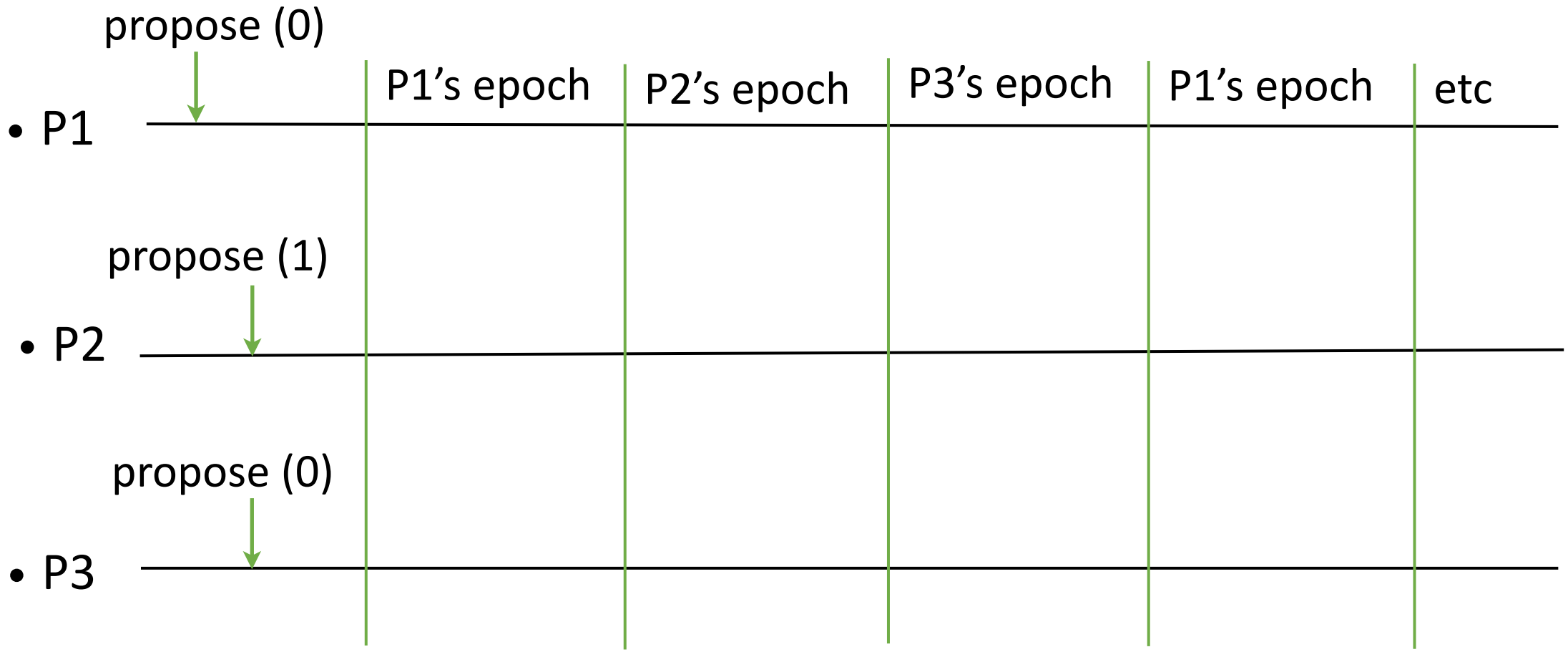
Before deciding, the leader first stores the value in a quorum.

If the leader fails, the next leader retrieves the value from a quorum, and completes the decision of that value in the system.

Leader election

- Processes move through rounds.
- Simplifying assumption: Process pi is leader in every round k if k mod N = i.
- The processes alternate in the role of a **leader** until one of them succeeds in imposing a decision.

Fail-silent Consensus

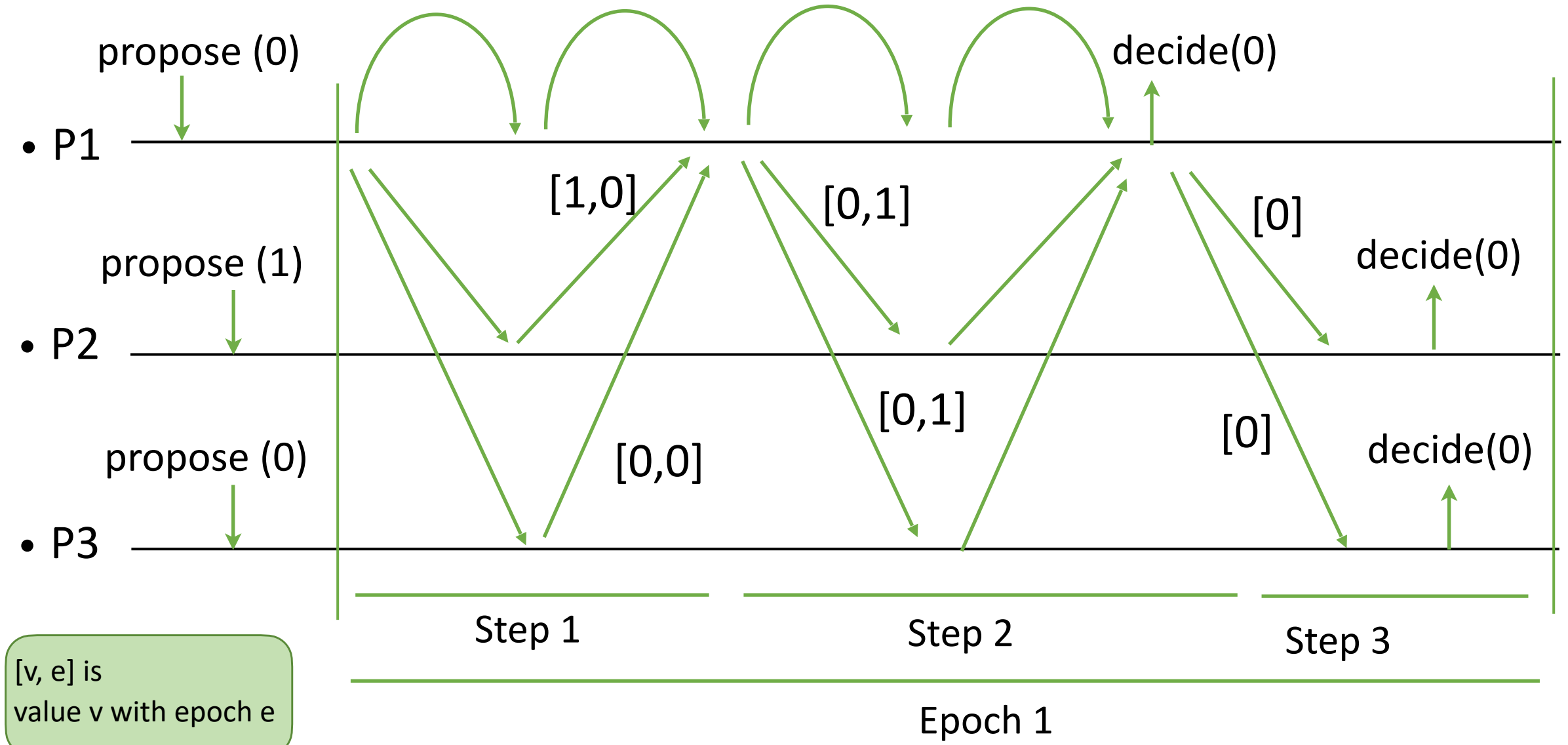


Fail-silent Consensus

To decide, a leader executes:

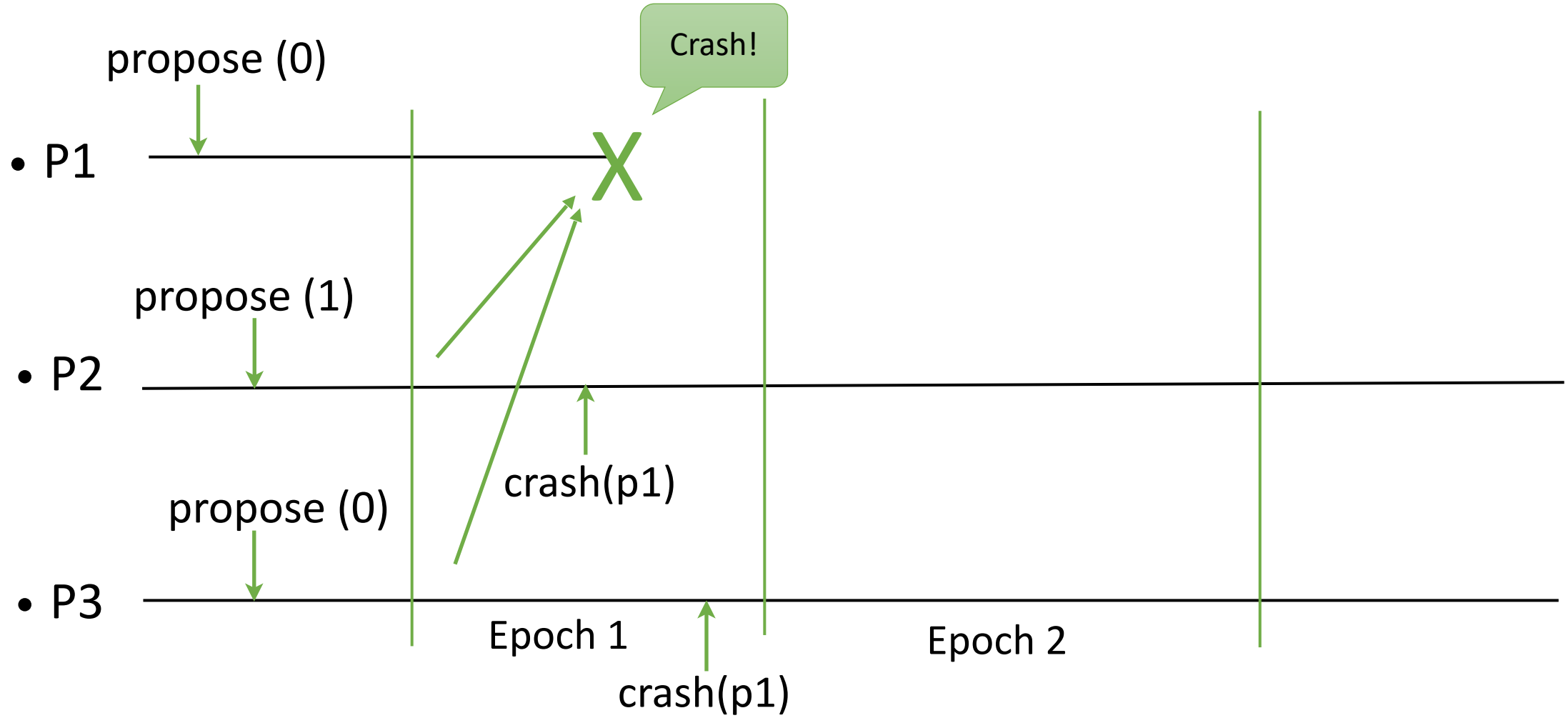
1. It reads the latest adopted value from a majority.
   The latest value according to the round number when the value has been adopted.
2. It imposes that value to a majority.
3. It decides and broadcasts the decision to all.

Fail-silent Consensus

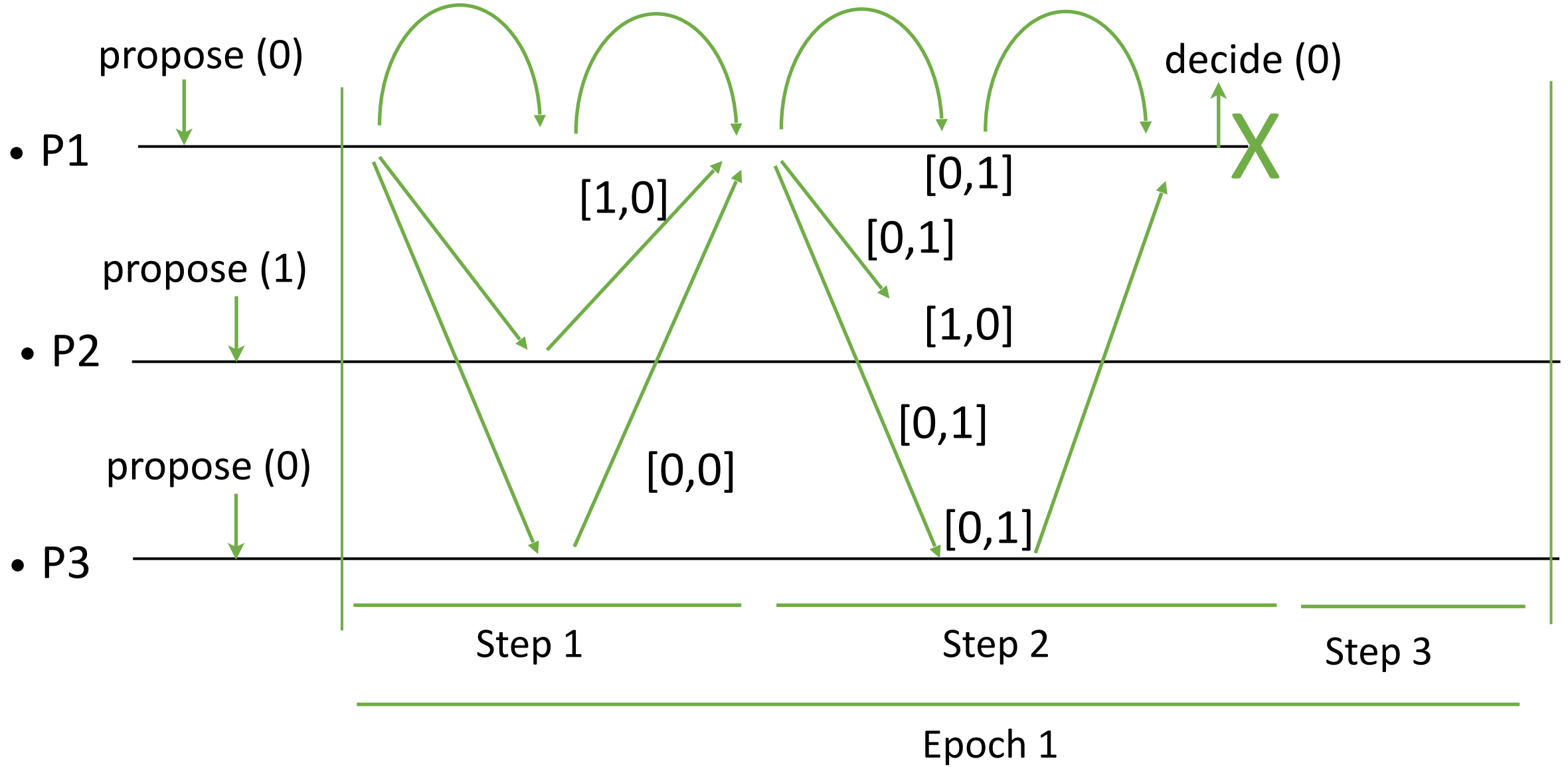


[v, e] is
value v with epoch e

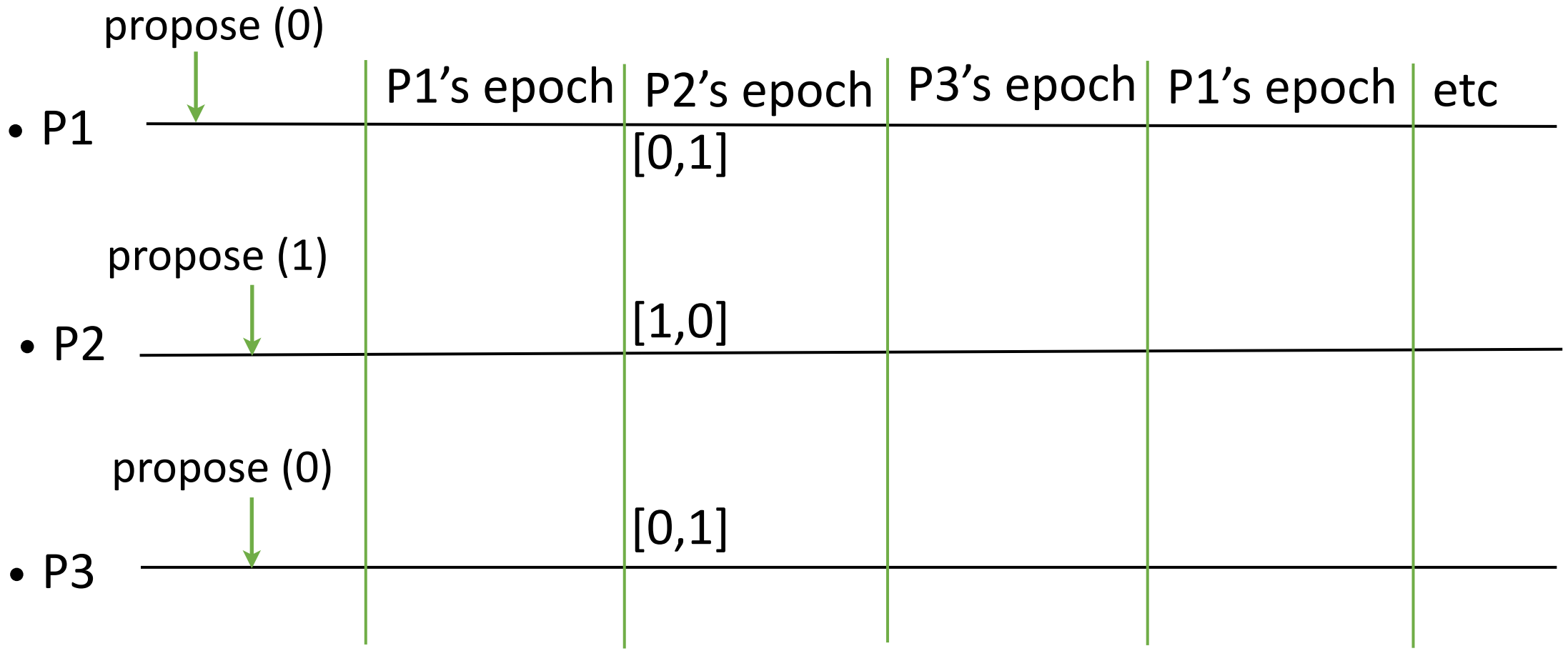
Fail-silent Consensus



Fail-silent Consensus

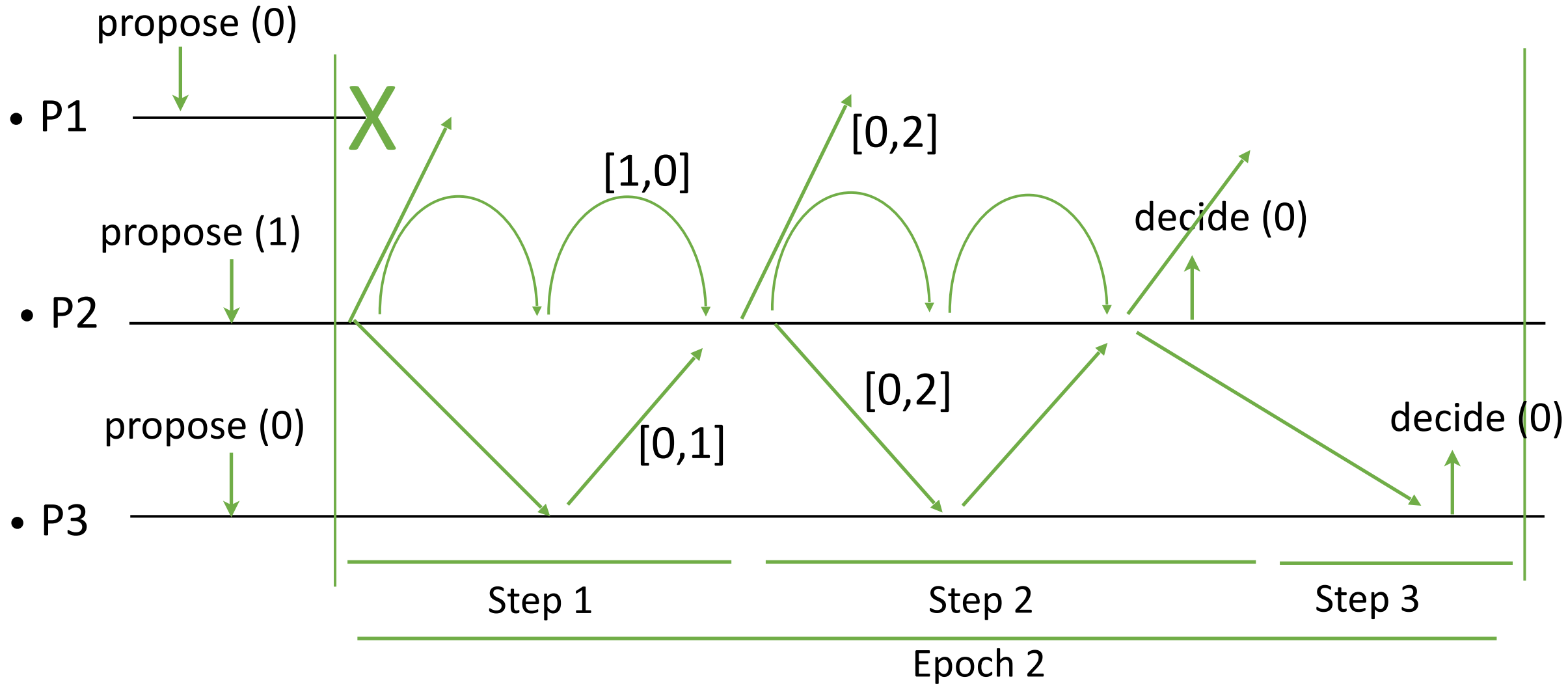


Fail-silent Consensus



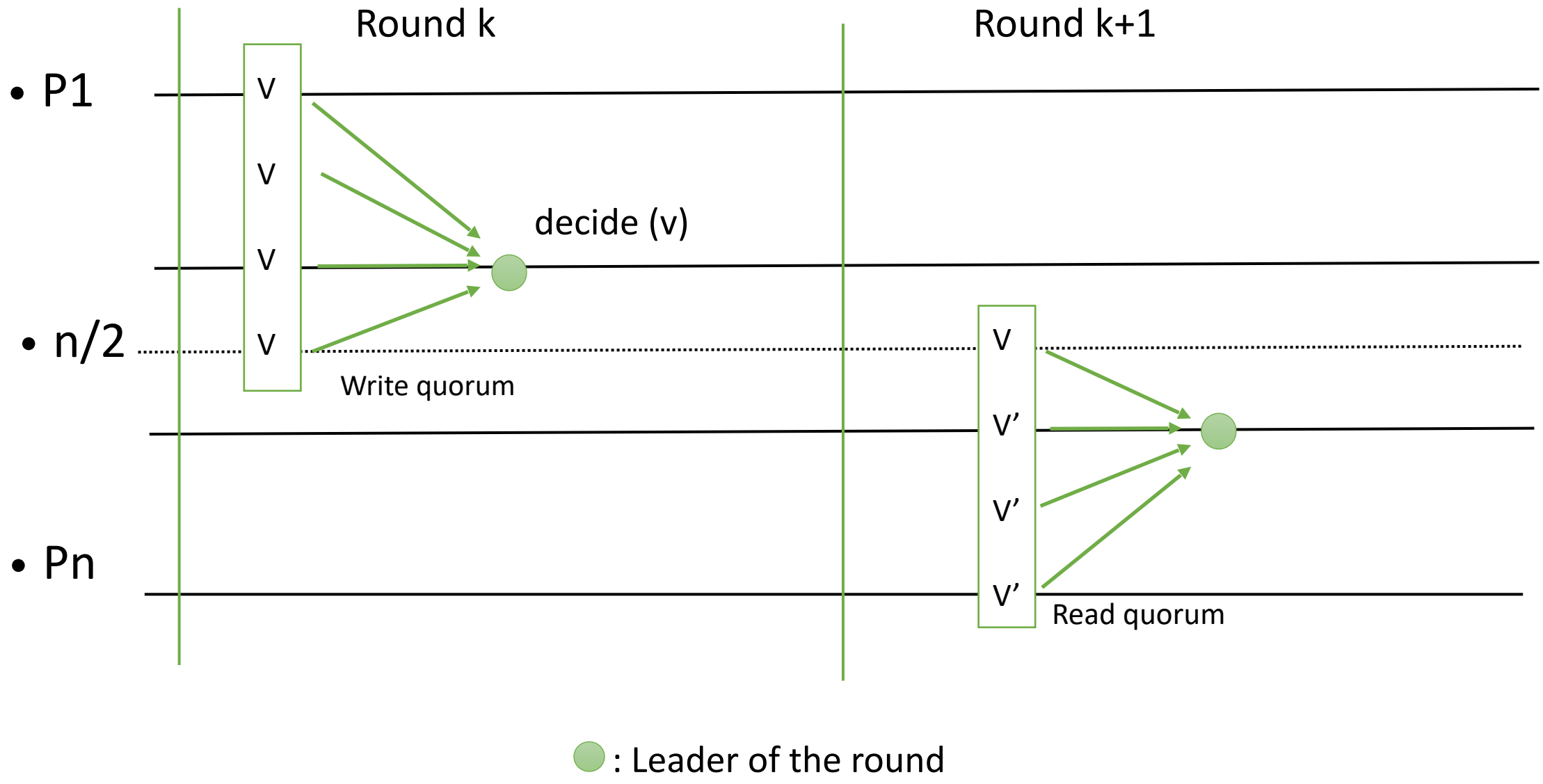
Consensus failed in round 1. It will continue to epoch 2.

Fail-silent Consensus



The process p2 adopts the already decided value 0 from a majority and imposes it.

Quorums intersect



Leaders

How are leaders elected and changed in rounds?

Eventual Leader Detector (Ω)

- Events :
  - Indication <Ω, trust(p)>

    Indicates that process p is trusted to be leader

- Properties :
  - ELD1 (Eventual accuracy): Eventually every correct process trusts some correct process.
  - ELD2 (Eventual agreement): Eventually no two correct processes trust different processes.

- The trusted leader of a process may change over time.
  The sequence of leaders might be different across processes.
  Eventually processes follow the same correct leader.

Partial synchrony and ◇P

- In an asynchronous system with processes prone to crash or Byzantine failures, deterministic algorithms cannot implement consensus [FLP].

- We assume a partially synchronous systems where we have an eventually perfect failure detector ◇P.

Monarchical Eventual Leader Detection

Implements:

EventualLeaderDetector, **instance** Ω.

Uses:

EventuallyPerfectFailureDetector, **instance** $\Diamond\mathcal{P}$.

upon event $\langle\ \Omega, Init\ \rangle$ **do**

$suspected := \emptyset$;

$leader := \perp$;

upon event $\langle\ \Diamond\mathcal{P}, Suspect \mid p\ \rangle$ **do**

$suspected := suspected \cup \{p\}$;

upon event $\langle\ \Diamond\mathcal{P}, Restore \mid p\ \rangle$ **do**

$suspected := suspected \setminus \{p\}$;

upon $leader \neq maxrank(\Pi \setminus suspected)$ **do**

$leader := maxrank(\Pi \setminus suspected)$;

trigger $\langle\ \Omega, Trust \mid leader\ \rangle$;

Leaders

How do we associate leaders to rounds?

Epoch-Change (ec)

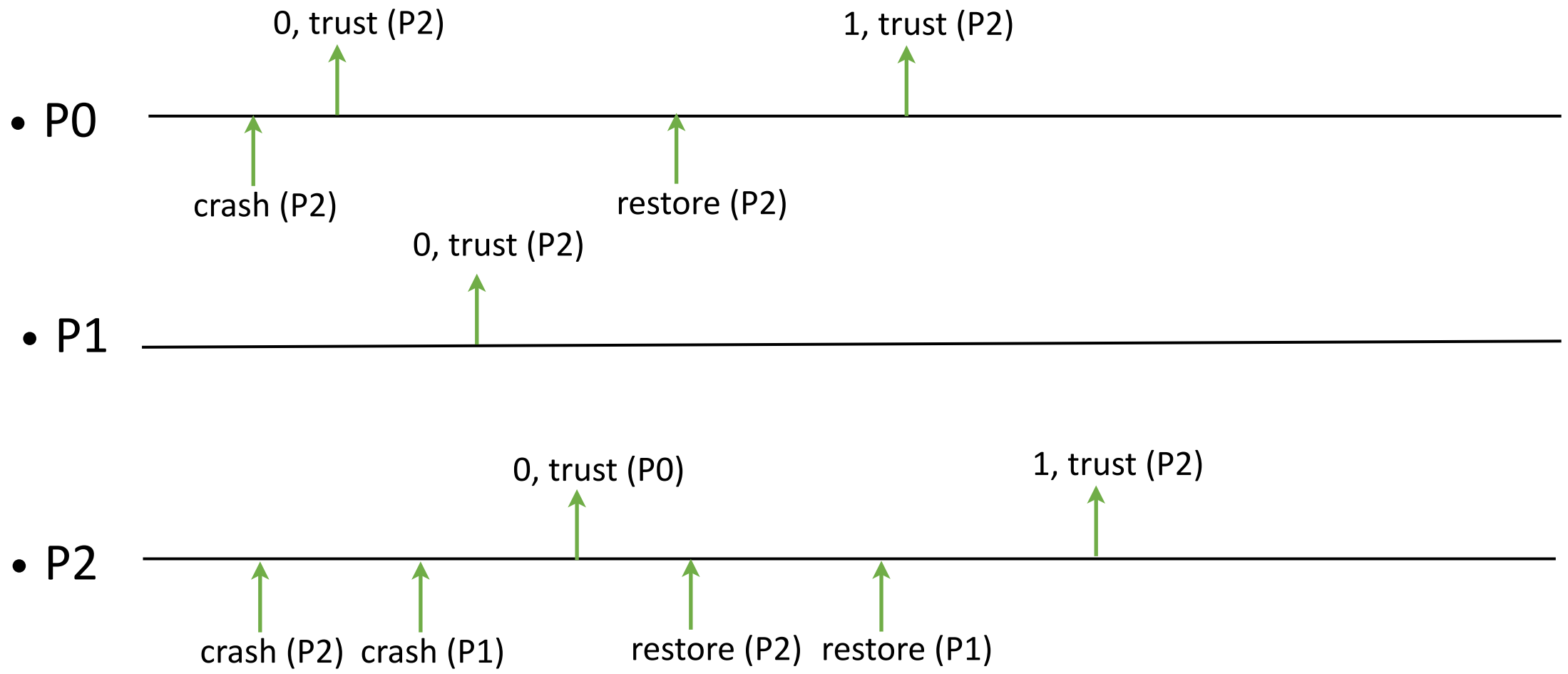
- Events:
  - Indication <ec, start-epoch(ts, L)>
    - Starts epoch (ts, L), timestamp ts and leader L
- Properties:
  - EC1 (Monotonicity):
    If a correct process starts epoch (ts, L) and later starts epoch (ts’, L'), then ts' > ts.
  - EC2 (Consistency):
    If a correct process starts epoch (ts, L) and another correct process starts epoch (ts, L'), then L = L'.
  - EC3 (Eventual Leadership):
    Eventually every correct process starts no further epoch; moreover, every correct process starts the same last epoch (ts, L), where L is a correct process.

Epoch-change, Monotonicity

- Use eventual leader detector (Ω)
- A locally increasing timestamp can provide monotonicity.
- Maintain current trusted leader and an increasing timestamp.

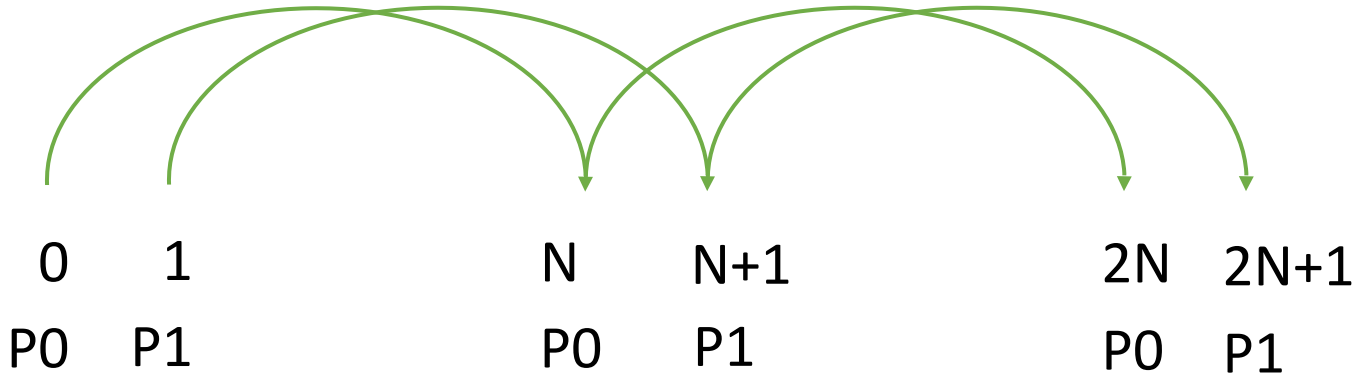
Epoch-change, Consistency

- However, “before eventually”, Ω does not guarantee the same leader or order of leaders at different processes. This can violate consistency.

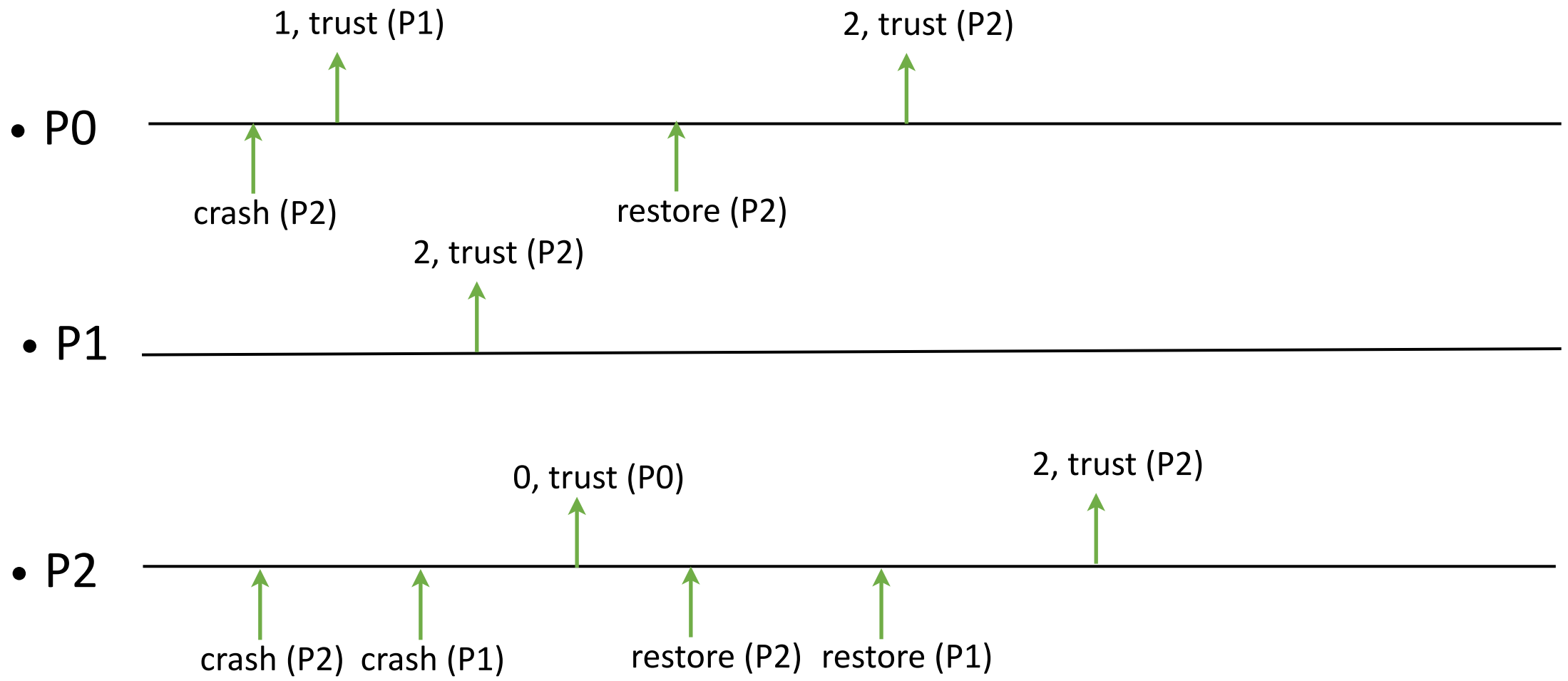


Epoch-change, Consistency

- Therefore, the timestamp domain is disjointly divided between processes.
- For a new leader, we jump to the next timestamp for that process.



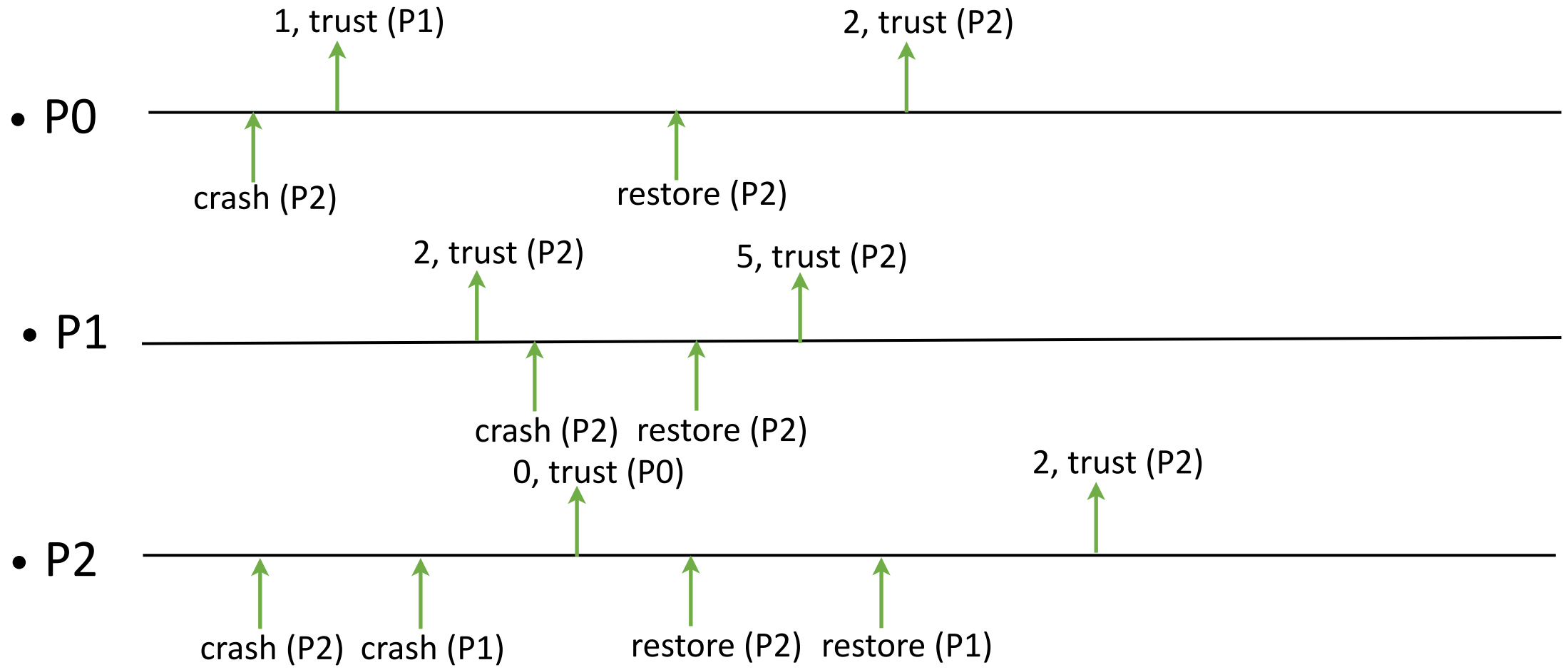
Epoch-change, Consistency



Epoch-change, Eventual leadership

- To have eventual leadership, the last leader should be installed with the *same timestamp* at all processes.

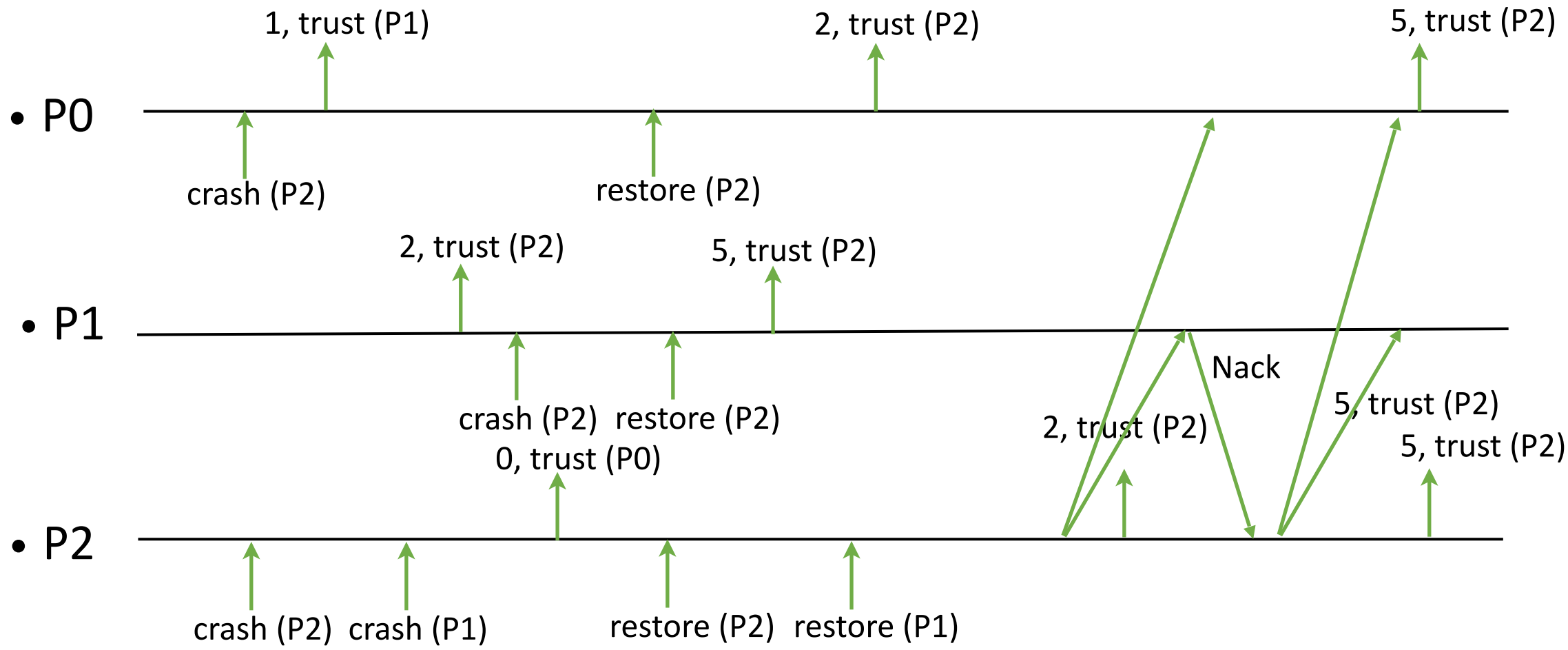
Epoch-change, Consistency



Epoch-change, Eventual leadership

- If a process finds himself to be the leader, he broadcasts himself as the leader with a timestamp.
- He might be rejected by a process that has a leader with a larger timestamp.
- The new leader repeats broadcasting with larger timestamps until he does not hear any rejections.

Epoch-change, Consistency



Epoch-Change Protocol

Implements:

EpochChange, **instance** *ec*.

Uses:

PerfectPointToPointLinks, **instance** *pl*;
BestEffortBroadcast, **instance** *beb*;
EventualLeaderDetector, **instance** Ω.

upon event $\langle$ *ec*, *Init* $\rangle$ **do**

$trusted := \ell_0$;
$lastts := 0$;
$ts := rank(self)$;

trusted is the latest leader that Ω suggested.
lastts is the last timestamp that was accepted for the leader.
ts is the timestamp that this process wants to impose on others (when Ω suggests her as the leader.)

Epoch-Change Protocol

upon event $\langle\ \Omega, \mathit{Trust} \mid p\ \rangle$ **do**
    $\mathit{trusted} := p$;
    if $p = \mathit{self}$ **then**
        *ts := (lastts/N + 1)*N + rank(self)*
        trigger $\langle\ \mathit{beb}, \mathit{Broadcast} \mid [\text{NEWEPOCH}, ts]\ \rangle$;

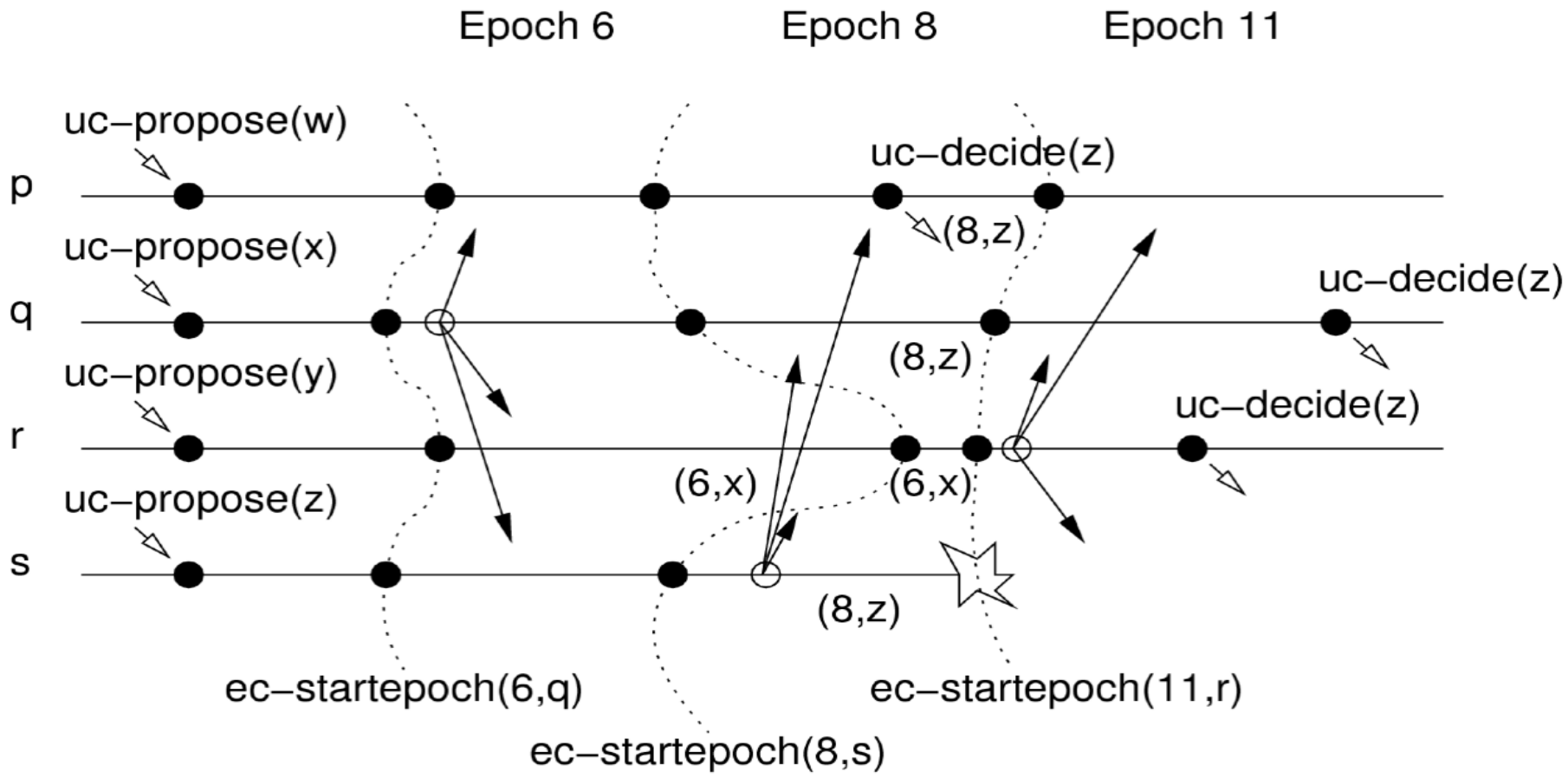
upon event $\langle\ \mathit{beb}, \mathit{Deliver} \mid \ell, [\text{NEWEPOCH}, \mathit{newts}]\ \rangle$ **do** **where** *newts ≠ lastts*
    if $\ell = \mathit{trusted} \wedge \mathit{newts} > \mathit{lastts}$ **then**
        $\mathit{lastts} := \mathit{newts}$;
        trigger $\langle\ \mathit{ec}, \mathit{StartEpoch} \mid \mathit{newts}, \ell\ \rangle$;
    else
        trigger $\langle\ \mathit{pl}, \mathit{Send} \mid \ell, [\text{NACK}]\ \rangle$;

upon event $\langle\ \mathit{pl}, \mathit{Deliver} \mid p, [\text{NACK}]\ \rangle$ **do**
    if $\mathit{trusted} = \mathit{self}$ **then**
        $ts := ts + N$;
        trigger $\langle\ \mathit{beb}, \mathit{Broadcast} \mid [\text{NEWEPOCH}, ts]\ \rangle$;

Eventual leadership

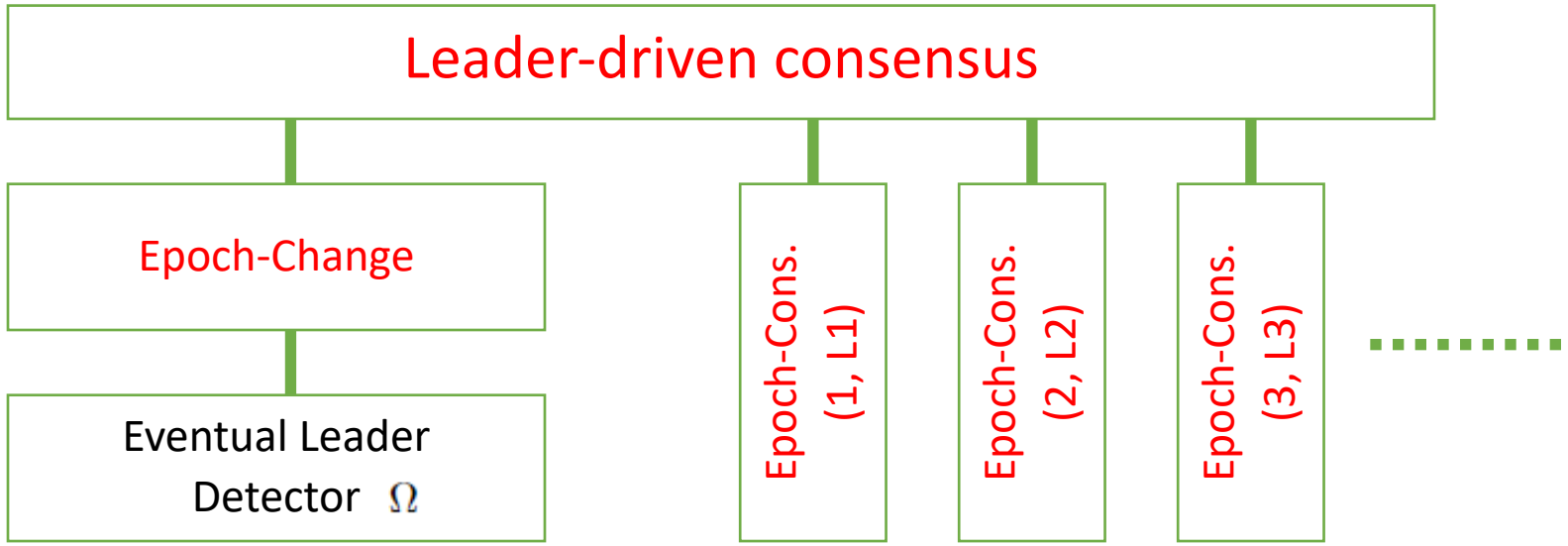
Without eventual accuracy, eventual leadership is not guaranteed:
Every leader may be suspected infinitely often.

Change of epochs



- Every process (p, q, r, s) uc-proposes a value
- Epoch 6 has leader q
  - q ep-proposes x, but only r receives it before epoch aborts
  - r now has state (6,x)
- Epoch 8 has leader s
  - s ep-proposes z, processes p, q, s receive it
  - only p ep-decides(z); then s crashes
- Epoch 11 has leader r,
  - It retrieves the value z and and ep-decides(z)

Leader-driven consensus



- Leader-driven consensus invokes
  - One instance of Epoch-Change (invokes Omega)
  - Multiple instances of Epoch Consensus
    - Identified by the epoch number and a designated leader

Uniform Consensus (uc)

- Events :
  - Request <uc, propose(v)>

    Proposes value v for consensus
  - Indication <uc, decide(v)>

    Outputs a decided value v of consensus

- Properties :
  - UC1 (Termination): Every correct process eventually decides.
  - UC2 (Validity): Any decided value has been proposed by some process.
  - UC3 (Integrity): No process decides twice.
  - UC4 (Uniform Agreement): No two processes* decide differently.

* both correct or incorrect

Epoch-Consensus (ep)

Associated with timestamp ts and leader L

Events:

- Request <ep, propose(v)>

  Proposes v for epoch consensus (executed by leader only)

- Indication <ep, decide(v)>

  Outputs decided value v for epoch consensus

- Request <ep, abort>

  Aborts this epoch consensus

- Indication <ep, aborted(s)>

  Signals that this epoch consensus has completed the abort and returns state s

Leader-driven consensus

Implements uc

uses ec, ep (multiple instances)

upon <init> **do**

val := ⊥; proposed := false; decided := false

(ets, L) := (0, L0); (newts, newL) := (0, ⊥)

Initialize Epoch Consensus instance ep[0] with timestamp 0 and leader L0

upon <uc, propose(v)> **do**

val := v

upon <ec, start-epoch(newts', newL')> **do**

(newts,newL) := (newts',newL')

trigger <ep[ets], abort>

Switching from an epoch to the next

upon <ep[ets], aborted(s)> **do**

(ets,L) := (newts,newL)

Initialize Epoch Consensus instance ep[ets] with timestamp ets, leader L, and state s

proposed := false

Leader-driven consensus

upon L = self $\wedge$ val $\neq \perp \wedge \neg$proposed **do**

proposed := true

trigger <ep[ets], propose(val)>

upon <ep[ets], decide(v)> **do**

if ¬decided **then**

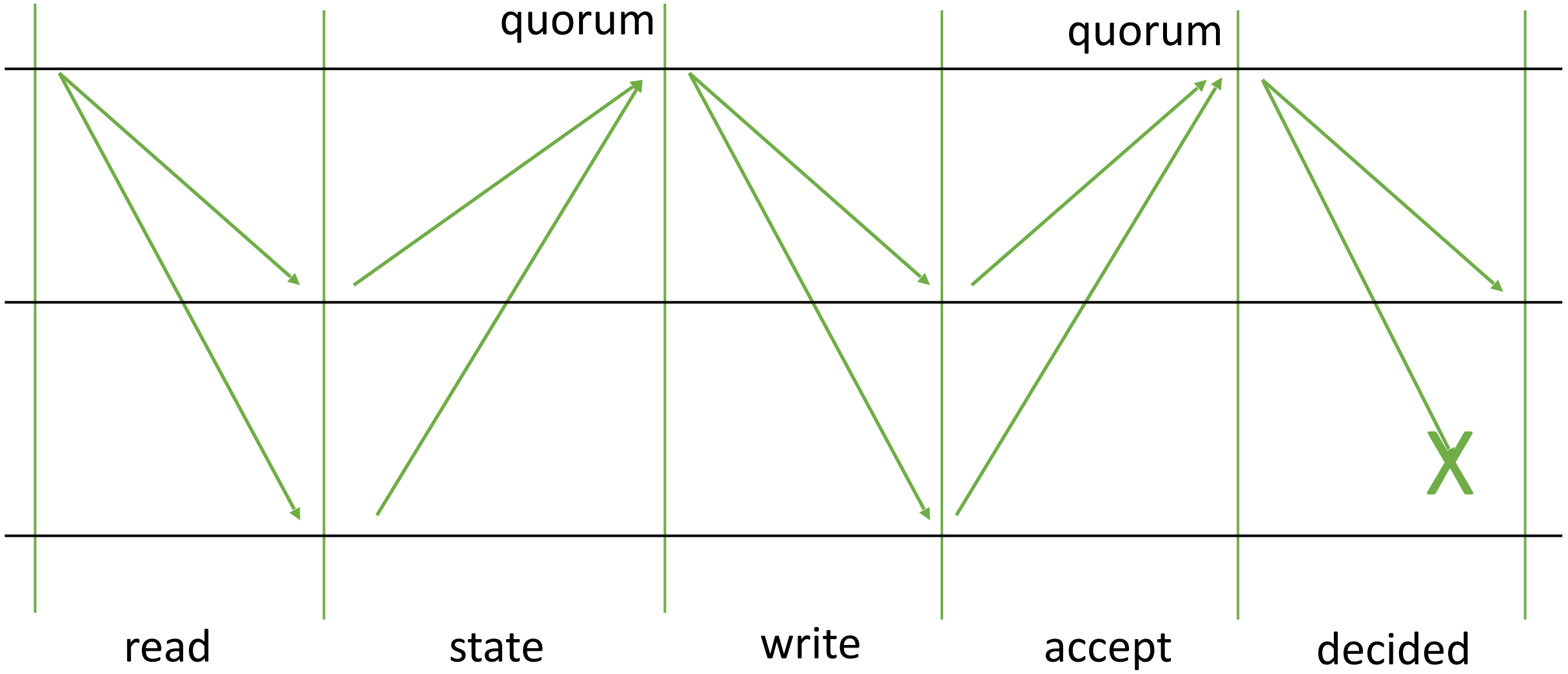
decided := true

trigger <decide(v)>

Implementing epoch consensus

- Read/write epoch consensus algorithm
  - Analogous to replicated implementation of a shared single-writer register
- State consists of a timestamp/value pair
- Leader reads state and looks for a value
  - Chooses value with highest timestamp
  - If no value found, takes its own proposal
  - Writes the chosen value
- Decide once a quorum of processes (> N/2) write the value and acknowledge

Epoch Consensus Steps



Read/write epoch consensus

Implements ep, uses pl, beb when N > 2f with timestamp ets and leader L

upon <ep, init(valts',val')> **do**
    proposal := ⊥; states := $[\perp]^N$; accepted := 0
    (valts, val) := (valts', val')

upon <ep, propose(v)> **do**
    proposal := v
    trigger <beb, broadcast(Read)>

upon <beb, deliver(L, Read)> **do**
    trigger <pl, Send(L, State(valts, val))>

upon <pl, deliver(p, State(ts, v))> **do**
    states[p] := (ts,v)

upon #(states) > N/2 **do**
    (ts,v) := highest(states); states := $[\perp]^N$
    if v ≠⊥ then proposal := v
    trigger <beb, broadcast(Write(proposal))>

Read/write epoch consensus

upon <beb, deliver(L, Write(v))> **do**
    (valts,val) := (ets,v)
    trigger <pl, Send(L, Accept)>

upon <pl, deliver(q, Accept)> **do**
    accepted := accepted + 1

upon accepted > N/2 **do**
    accepted := 0
    trigger <beb, broadcast(Decided(proposal))>

upon <pl, deliver(L, Decided(v))> **do**
    trigger <ep, decide(v)>

upon <ep, abort> **do**
    trigger <ep, aborted(valts,val)>

Epoch-Consensus (ep)

Associated with timestamp ts and leader L

Events:

- Request <ep, propose(v)>

  Proposes v for epoch consensus (executed by leader only)

- Request <ep, abort>

  Aborts this epoch consensus

- Indication <ep, decide(v)>

  Outputs decided value v for epoch consensus

- Indication <ep, aborted(s)>

  Signals that this epoch consensus has completed the abort and returns state s

Epoch-Consensus (ep)

- Properties
  - EP1 (Validity):
    If a correct process decides v in an epoch ts, then v was proposed by the leader of some epoch consensus with ts' ≤ ts.
  - EP2 (Uniform Agreement):
    No two [correct*] processes decide differently. *for Byzantine epoch consensus
  - EP3 (Integrity):
    A correct process decides at most once.
  - EP4 (Lock-in):
    If a process decides v in epoch ts' < ts, no process decides a value different from v in epoch ts.
  - EP5 (Termination):
    If the leader L is correct, has proposed a value and no process aborts, then every correct process eventually decides.
  - EP6 (Abort behavior):
    When a correct process aborts, then it eventually completes the abort; plus, a correct process completes an aborts only if it has been aborted before.
- Every process must run a well-formed sequence of epoch consensus instances, only one instance of epoch consensus at a time. Give state from previous (aborted) instance to next instance. Associated timestamps monotonically increasing.

Correctness

- Validity (EP1)
  - The decided value either comes from a State message or from the leader L itself.
  - The value of a State message has been (inductively) written by some leader.
- Uniform Agreement (EP2)
  - Immediate from the fact that every process decides only the value of the Decided message. The leader sends the Decided message only once.
- Integrity (EP3)
  - The Decided message is sent at most once: when it is sent, the accepted variable is set to zero and cannot get more than N/2 again.
- Lock-in (EP4)
  - A write-quorum (> N/2) stored v before sending the Accept message in previous epoch ts'.
  - Processes passed it in the state to subsequent epochs to ts.
  - Then, L reads v in at least one State message from a read-quorum (> N/2).
- Termination (EP5)
  - A majority is correct, and the correct leader can complete all the steps.
- Abort behavior (EP6)
  - Immediate from the algorithm.

Uniform Consensus (uc)

- Events :
  - Request <uc, Propose(v)>

    Proposes value v for consensus
  - Indication <uc, decide(v)>

    Outputs a decided value v of consensus

- Properties :
  - UC1 (Termination): Every correct process eventually decides.
  - UC2 (Validity): Any decided value has been proposed by some process.
  - UC3 (Integrity): No process decides twice.
  - UC4 (Uniform Agreement): No two processes* decide differently.

* both correct or faulty

Correctness

- Termination (UC1 / WBC1)
  - From EC3 (eventual leadership), EP5 (termination) and the algorithm (no redundant aborts)
- Validity (UC2) / Weak Validity (WBC2)
  - From EP1 (validity) and the algorithm (leaders store and propose user proposals).
- Integrity (UC3)
  - The decided variable in the algorithm.
- Uniform Agreement (UC4 / WBC4)
  - From the algorithm (decide only after decide from ep), EP2 (agreement), and EP4 (lock-in)

We will talk about WBC (Weak Byzantine Consensus) in future lectures.

Parts of slides adopted from C. Cachin, R. Guerraoui, L. Rodrigues